Quantifying Transient Storage and Hyporheic Exchange using Tracer Techniques and the OTIS Solute Transport Model

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Workshop Outline

I. Basic Theory on Solute Transport and Modeling

A. Intro/Overview

B. Tracer & Modeling 101: Mass Balance

C. Stream Transport Processes: the Advection-Dispersion Equation

D. Solving the Advection-Dispersion Equation: Numerical Methods

E. Stream Transport Processes II: Transient Storage

II. Use of OTIS for Quantifying Transient Storage — Uvas Creek Example

A. Steps 1&2: Initial Characterization of Transport w/o Modeling

B. Step 3: Initial OTIS Runs

C. Step 4: Automated Parameter Estimation (OTIS-P) — Advection-Dispersion

D. Step 5: OTIS-P — Advection-Dispersion-Storage

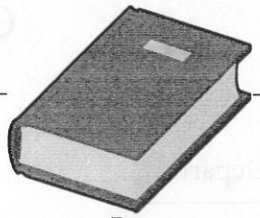
E. Boring, but important

Intro/Overview

Early USGS Tracer Applications

The Tracer-Dilution Method:
Kilpatrick & Cobb
TWRI Book 3, Ch. A16

Measurement of Discharge Using Tracers

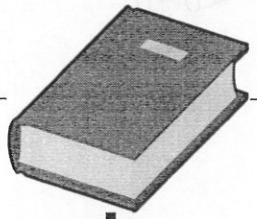


Synoptic Studies (Constituent Loads):
T-D method & Spatially-detailed sampling

K. Bencala, B. Kimball, D. McKnight

Time of Travel:
Kilpatrick & Wilson
TWRI Book 3, Ch. A9

Measurement of Time of Travel in Streams



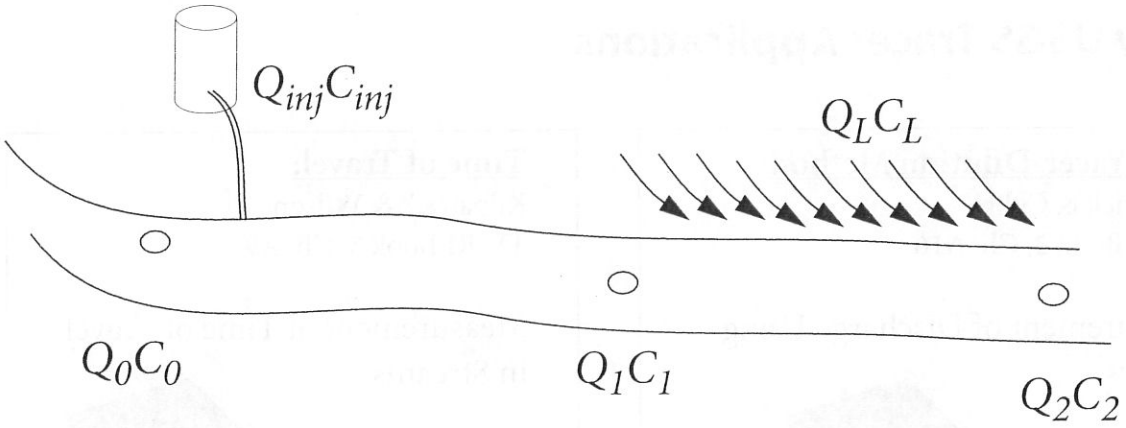
Small Stream Hydrodynamics, GW/SW Interaction; T. Storage

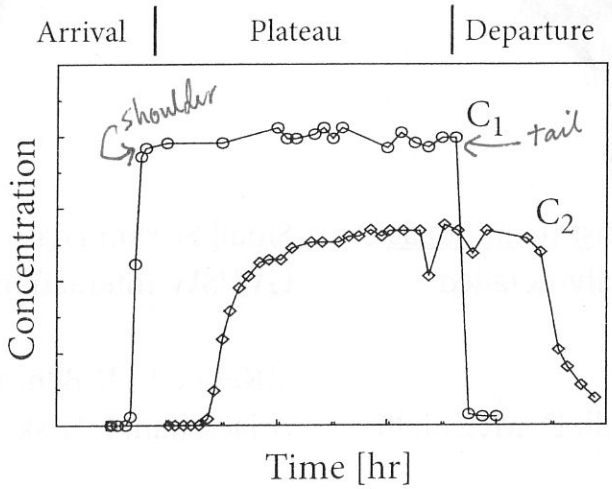
V.Kennedy, K.Bencala, R.Walters, A.Jackman, F.Triska, R.Rathbun

Focus of this Workshop

http://smig.usgs.gov/SMIG/reading_refs.html

Continuous Tracer Injection





Types of Tracers & Tracer Additions

Types of Tracers:

Ionic Tracers (Salts) — Low pH: LiCl, LiBr;
Neutral pH: NaBr, KBr, NaCl

Dyes — Rhodamine WT

Types of Tracers Additions:

Continuous, Constant Rate Injections — Tracer introduced at a constant rate and concentration using a tank & metering pump (see previous slide)

Slug/Pulse Injections — "Instantaneous" introduction of Tracer; typically a slug of dye (see Workshop Cover page).

Modeling Approaches

Forward, Predictive Mode —

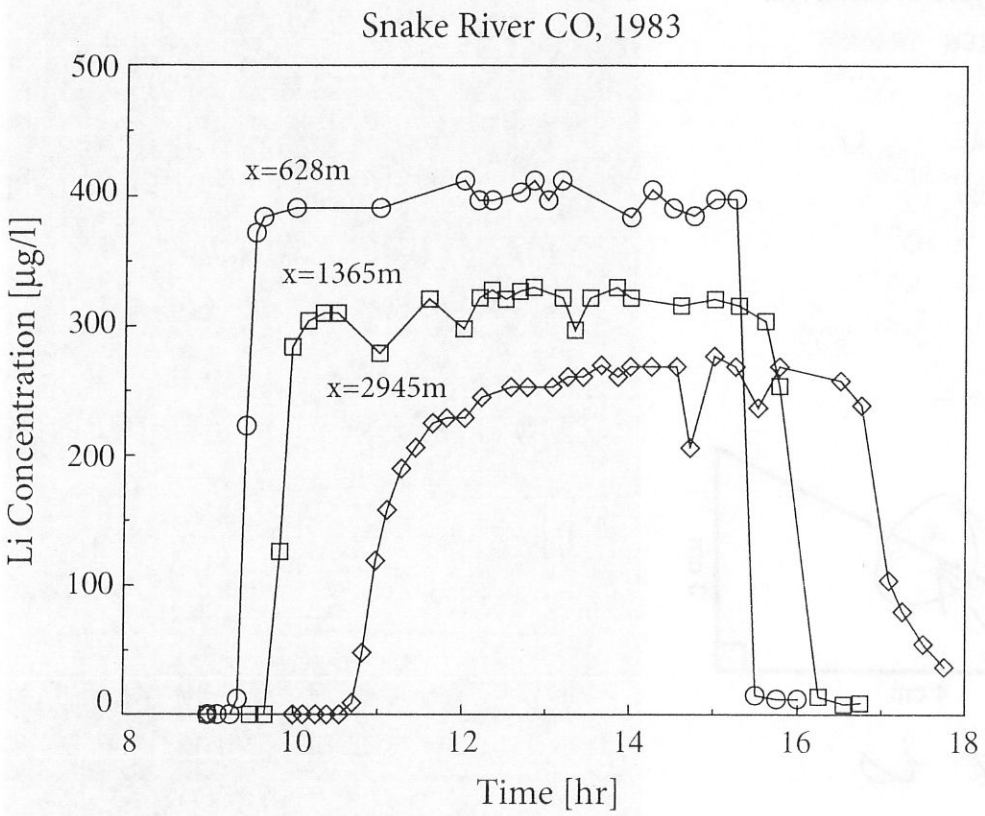
- Using a "Calibrated" model, predict concentration as a function of Time and Space
- Planning for Infrastructure (WWTPs) and Remediation

Inverse Mode / Parameter Estimation —

- Using observed data, estimate parameters that describe the physical and/or biogeochemical system
- Focus of this Workshop

Inverse Mode — Quantify Stream Hydrodynamics with Obs. Data

Downstream Conc. Profiles --> Transport Parameters



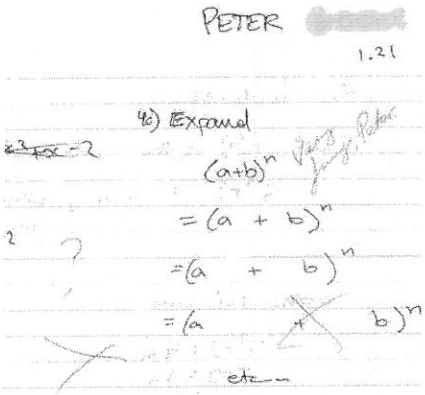
Magnitude --> Streamflow

Traveltime --> Velocity (Streamflow/Cross-Sectional Area)

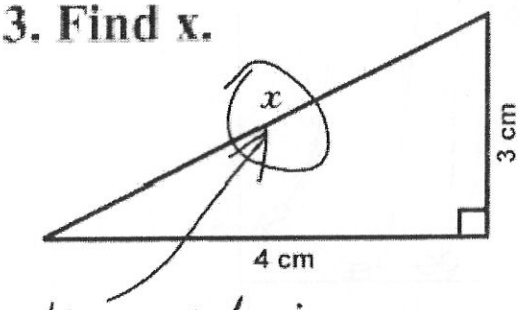
Shape --> Mixing (Dispersion & Transient Storage)

Tracer & Modeling 101: Mass Balance

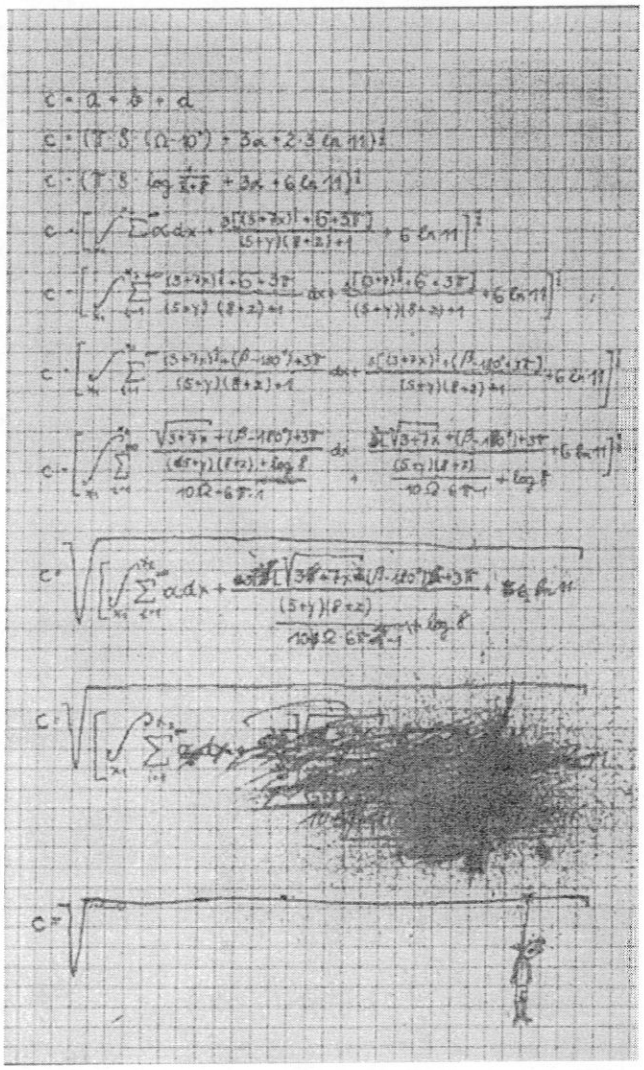
Future math majors of America:



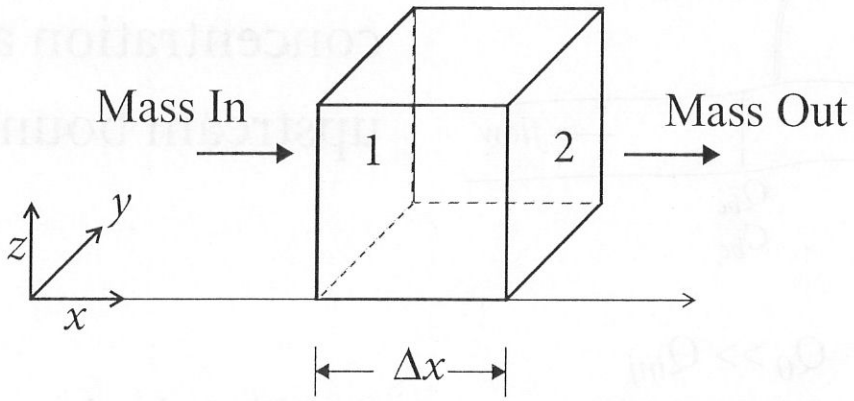
3. Find x.



Here it is



Mass Balance is our friend



Conservation of Mass:

$$Accumulation = \text{Mass In} - \text{Mass Out}$$

At steady-state (no change in time), Accumulation = 0,

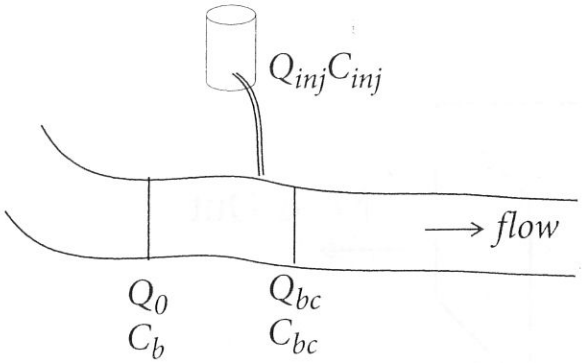
$$\text{Mass In} = \text{Mass Out}$$

"Mass Flow" — product of streamflow (Q) and conc. (C):

$$Q = \frac{volume}{time} = \frac{L^3}{T} \qquad C = \frac{mass}{volume} = \frac{M}{L^3}$$

$$\text{Mass Flow} = QC = \frac{mass}{time}$$

Continuous Tracer Injection—Mass Balance



What is the tracer concentration at the upstream boundary?

Assume: $Q_0 >> Q_{inj}$
instantaneous mixing w/ width & depth

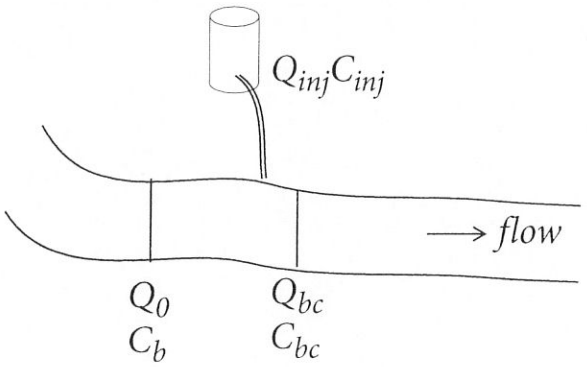
Conservation of Mass:

$$\text{Mass In} = \text{Mass Out}$$

$$QC_b + Q_{inj}C_{inj} = QC_{bc}$$

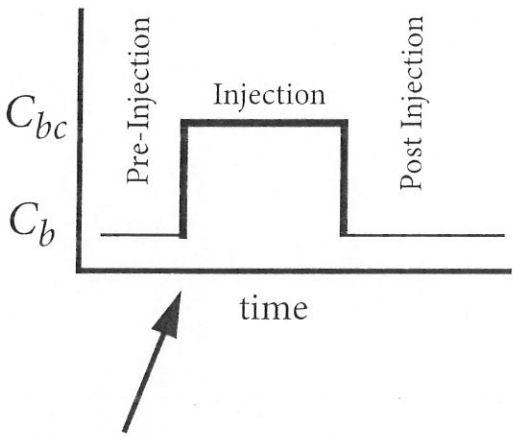
$$C_{bc} = C_b + \frac{Q_{inj}C_{inj}}{Q_{bc}} = \text{what was there + what you added}$$

Continuous Tracer Injection—Mass Balance



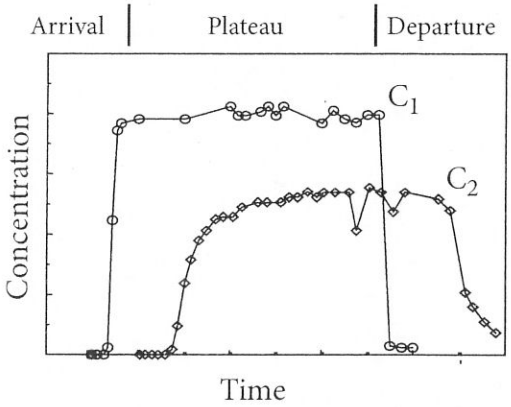
What is the tracer concentration at the upstream boundary?

Concentration at Injection (Boundary Condition)

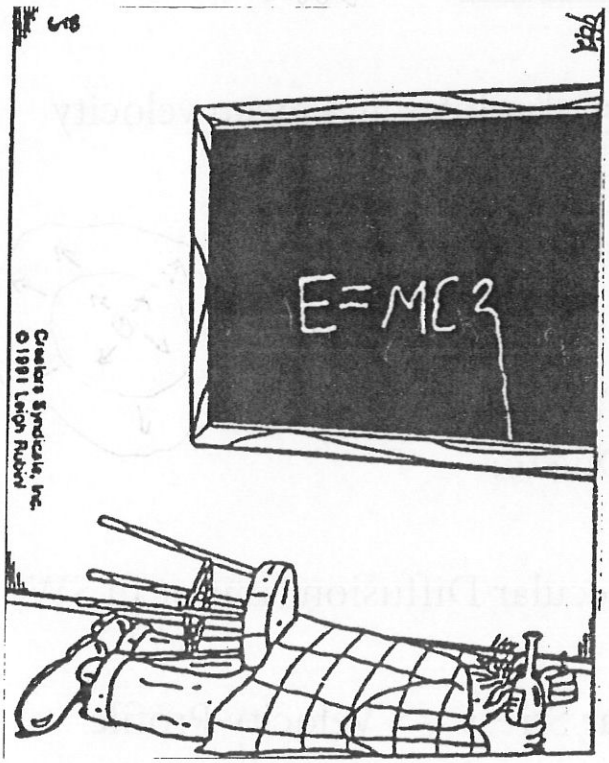


Rectangular (step) boundary condition

Downstream Concentration



Stream Transport Processes: the Advection-Dispersion Equation



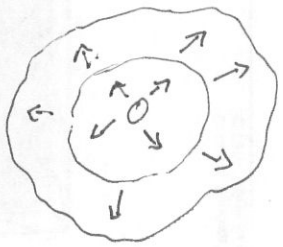
Early in his career, Einstein discovered the hazards of drinking and deriving

Quantifying Stream Transport Processes: The Advection-Dispersion Equation

Advection (Convection) — Downstream movement

Transport of solutes at a mean velocity

Dispersion — spreading out



Mixing Due to:

Molecular Diffusion (minor in SW)

Shear Stress --> Velocity Profile

Stream Transport Processes: Advection-Dispersion Eqn
Molecular Diffusion

Flux: Solute mass crossing unit area per unit time

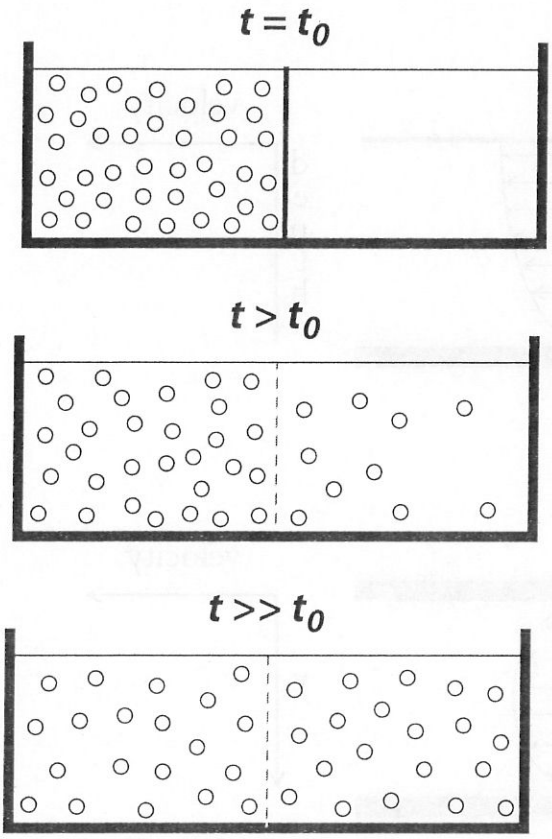


Fig. 1. Adolph Fick, pioneer in the application of physics, chemistry, and mathematics to biological systems (Fick, 1903; Frontispiece).

Fick's Law: Flux is proportional to the concentration gradient:

$$q = -D_m \frac{\partial C}{\partial x}$$

Stream Transport Processes: Advection-Dispersion Eqn
Velocity variations due to shear stress

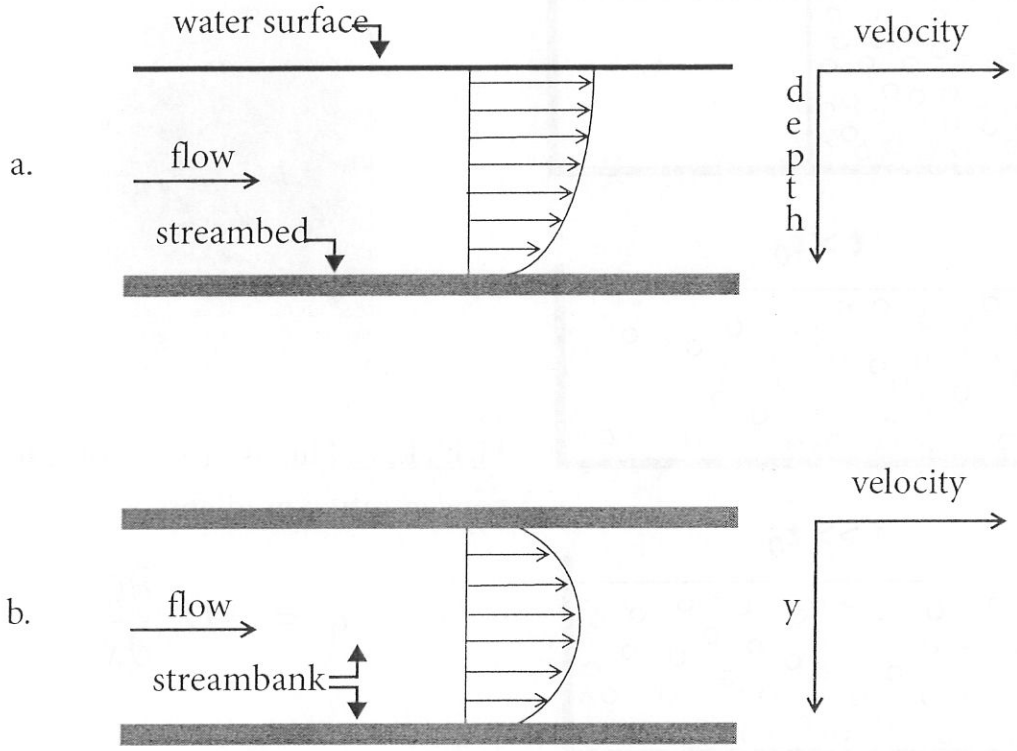


Figure 1:. a) Vertical velocity profile in a stream. b) Horizontal velocity profile in a stream. (Runkel and Bencala, 1995).

Stream Transport Processes: Advection-Dispersion Eqn
Dispersion

Small Scale (Diffusion) & Large Scale (Shear Stress) Mixing

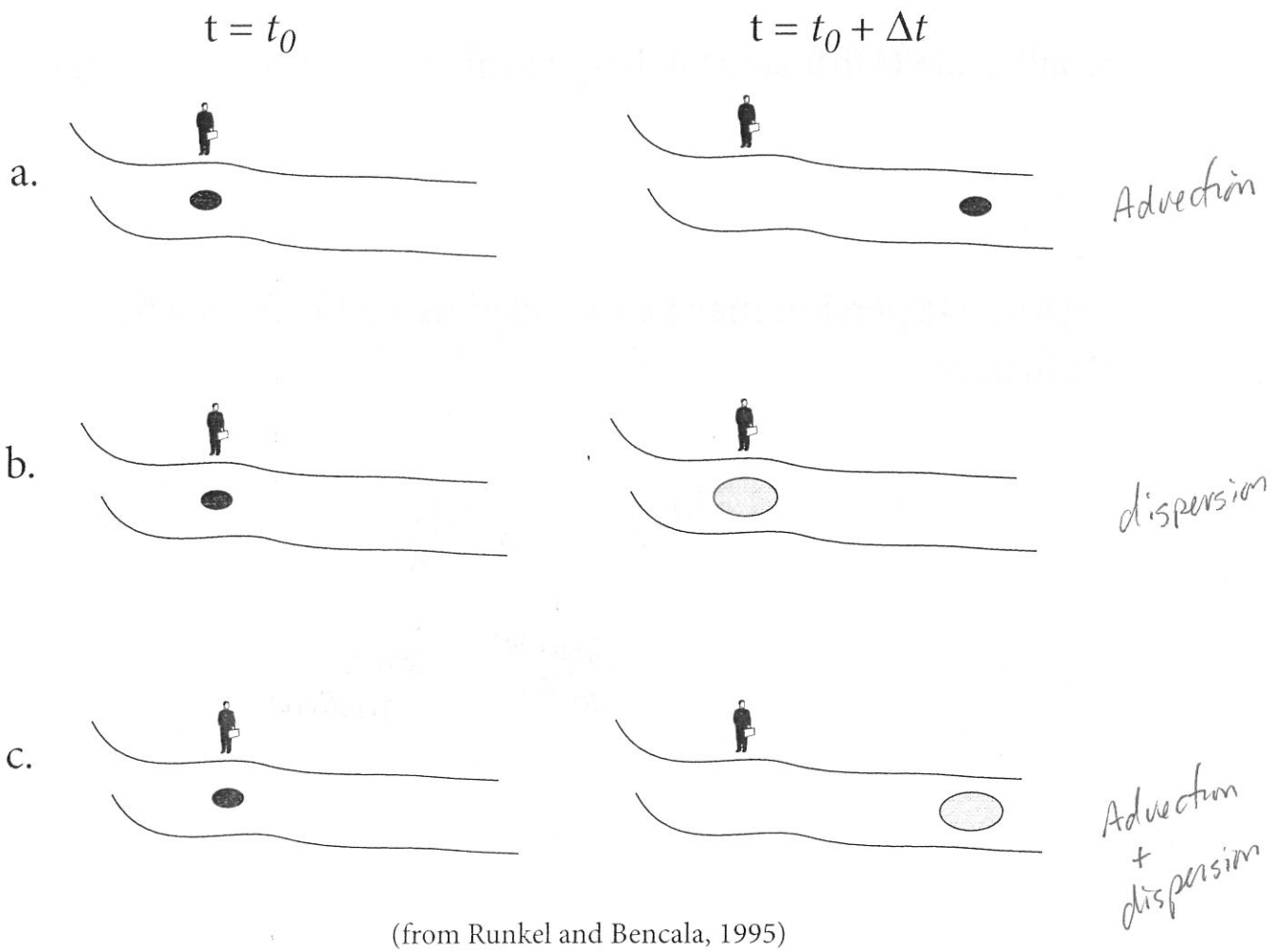
Taylor: Dispersion may be modeled using Fick's Law for Diffusion:

$$q_{disp} = -D_L \frac{\partial C}{\partial x} \qquad (2)$$

dispersion coeff.

conc. gradient

Stream Transport Processes: Advection-Dispersion Eqn
Combined Effects of Advection & Dispersion (panel c)



(from Runkel and Bencala, 1995)

Deep thoughts on Modeling, Math, Science and Computers

I think there is a world market for about five computers
- Thomas J. Watson, IBM Chair, 1943

The primary purpose of the DATA statement is to give names to constants; instead of referring to pi as 3.141592653589793, the variable PI can be given that value with a DATA statement and used instead of the longer form of the constant. This also simplifies modifying the program, *should the value of pi change.*

-- Fortran manual for Xerox Computers

2 + 2 = 4, except for large values of 2.

```
There are two major products that come
out of Berkeley: LSD and UNIX.  We don't
believe this to be a coincidence
                    -- Jeremy S. Anderson
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The models may be agreeing now simply because they're all tending to do the same thing wrong - Robert Cess

We can make the entire calculation look slightly less juvenile by introducing an obscure notation - Philip Pechukas

"We have avoided **repetition** wherever possible to avoid **repetition** and to be more concise. This suggestion (avoiding **repetition**) has been applied globally to the paper" - Author's response to reviewer comment that their manuscript contained redundant/repetitive information

Never make predictions, especially about the future — Casey Stengel

The climate modelers have been 'cheating' for so long, it's almost become respectable - Richard Kerr

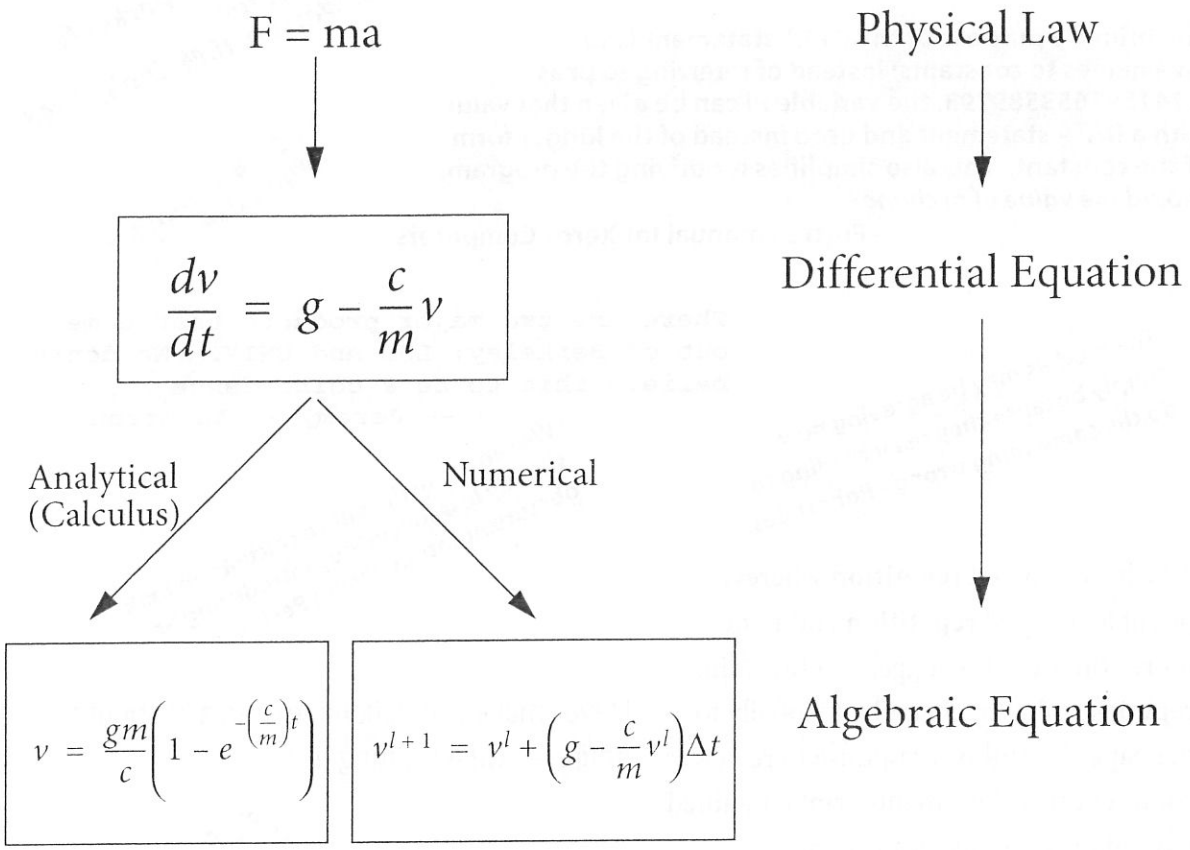
The more you learn, the more you understand that you don't understand very much.
-- Anonymous quote on climate

Engineers, particularly civil engineers, have been the principle developers of the field of water quality modeling. This might seem odd considering that civil engineers are most commonly associated with areas such as structural design and transportation. In fact, rather than being environmentally oriented, civil engineers are sometimes stereotyped as hard-hatted individuals who toss drink containers out the window of their pickup truck while simultaneously trying to hit any small mammal that crosses their path - Steve Chapra

"Drugs have taught an entire generation of American kids the metric system." - P.J. O'Rourke

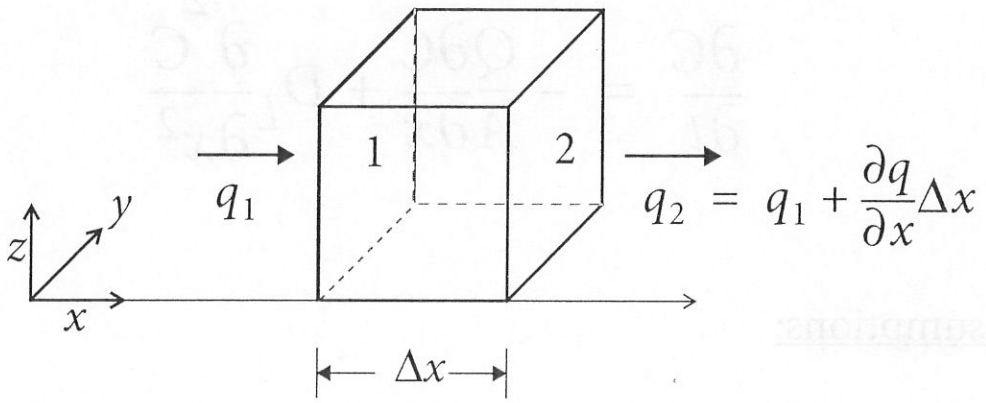
For every problem, there is one solution which is simple, neat and wrong - H.L. Mencken

Stream Transport Processes: Advection-Dispersion Eqn
Modeling: Physical Law --> Algebraic Equation



(Chapra & Canale, 1988)

Stream Transport Processes: Advection-Dispersion Eqn
Conservation of Mass



$$Accumulation = Mass\ In - Mass\ Out \quad (3)$$

Δ conc. vs. time

$$V\frac{\partial C}{\partial t} = q_1 - q_2 \quad (4)$$

where the fluxes (q) are developed for advection & dispersion as in Runkel and Bencala (1995)

Stream Transport Processes: Constant Parameter, 1-Dimensional Advection-Dispersion Equation

$$\frac{\partial C}{\partial t} = -\frac{Q}{A}\frac{\partial C}{\partial x} + D_L\frac{\partial^2 C}{\partial x^2} \qquad (5)$$

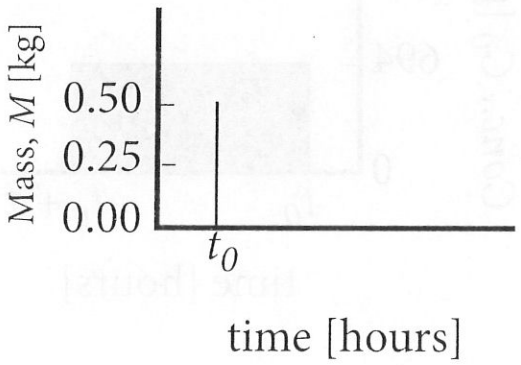
Assumptions:

- 1-Dimensional System
  (well-mixed in across A — w/ width & depth) simple canal
- Constant Parameters (A, Q & D_L spatially constant)
- Constant Volume (Steady Flow Regime)
- Conservative Solute - no biogeochemical reactions

Stream Transport Processes:
Analytical Solutions to the Advection/Dispersion Equation

Boundary Condition:

Instantaneous Source (e.g. Spill, Dye Slug)



Solution:

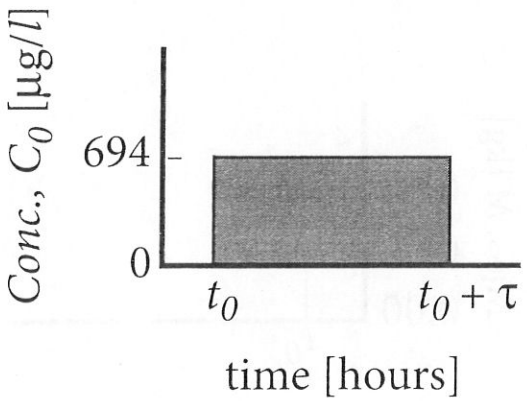
$$C(x,t) = \frac{M}{2A\sqrt{\pi D_L t}} exp\left[\frac{-(x-Ut)^2}{4D_L t}\right] \quad (6)$$

M: mass; U: discharge; D_L: dispersion coeff

Stream Transport Processes:
Analytical Solutions to the Advection/Dispersion Equation

Boundary Condition:

Rectangular Source (e.g. Constant-Rate Tracer Injection)



Solution:

(7)

for $t < \tau$

$$C(x,t) = \frac{C_0}{2}\left[erfc\left(\frac{x - Ut}{2\sqrt{D_L t}}\right) + exp\left(\frac{Ux}{D_L}\right)erfc\left(\frac{x + Ut}{2\sqrt{D_L t}}\right)\right]$$

(8)

for $t > \tau$

$$C(x,t) = \frac{C_0}{2}\left[erfc\left(\frac{x - Ut}{2\sqrt{D_L t}}\right) - erfc\left(\frac{x - U(t-\tau)}{2\sqrt{D_L (t-\tau)}}\right)\right.$$

$$\left. + exp\left(\frac{Ux}{D_L}\right)\left\{erfc\left(\frac{x + Ut}{2\sqrt{D_L t}}\right) - erfc\left(\frac{x + U(t-\tau)}{2\sqrt{D_L (t-\tau)}}\right)\right\}\right]$$

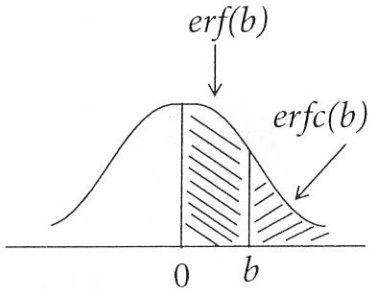
For solution w/ First-Order Decay, see: Runkel, J.Envir. Eng. 122(9), 1996.

Stream Transport Processes: Advection-Dispersion Eqn
The Error Function

$$erf(b) = \frac{2}{\sqrt{\Pi}} \int_0^b e^{-\xi^2} d\xi \qquad (9)$$

$$erfc(b) = 1 - erf(b) \qquad (10)$$

b	erf(b)	erfc(b)	b	erf(b)	erfc(b)
0.00	0.000000	1.000000	1.00	0.842701	0.157299
0.05	0.056372	0.943628	1.05	0.862436	0.137564
0.10	0.112463	0.887537	1.10	0.880205	0.119795
0.15	0.167996	0.832004	1.15	0.896124	0.103876
0.20	0.222703	0.777297	1.20	0.910314	0.089686
0.25	0.276326	0.723674	1.25	0.922900	0.077100
0.30	0.328627	0.671373	1.30	0.934008	0.065992
0.35	0.379382	0.620618	1.35	0.943762	0.056238
0.40	0.428392	0.571608	1.40	0.952285	0.047715
0.45	0.475482	0.524518	1.45	0.959695	0.040305
0.50	0.520500	0.479500	1.50	0.966105	0.033895
0.55	0.563323	0.436677	1.55	0.971623	0.028377
0.60	0.603856	0.396144	1.60	0.976348	0.023652
0.65	0.642029	0.357971	1.65	0.980376	0.019624
0.70	0.677801	0.322199	1.70	0.983790	0.016210
0.75	0.711155	0.288845	1.75	0.986672	0.013328
0.80	0.742101	0.257899	1.80	0.989091	0.010909
0.85	0.770668	0.229332	1.85	0.991111	0.008889
0.90	0.796908	0.203092	1.90	0.992790	0.007210
0.95	0.820891	0.179109	1.95	0.994179	0.005821



$$efc(0) = 0$$

$$efc(\infty) = 1$$

$$erf(-b) = -erf(b)$$

Stream Transport Processes: Advection-Dispersion Eqn Analytical Solution Example, Contaminant Release

Scenarios:

a) Spill scenario is viewed as an instantaneous release of 0.5 kg of mass.

b) Leak scenario; mass is released continuously over a two hour period. Concentration in the vicinity of the release is 694 μg/*l*.

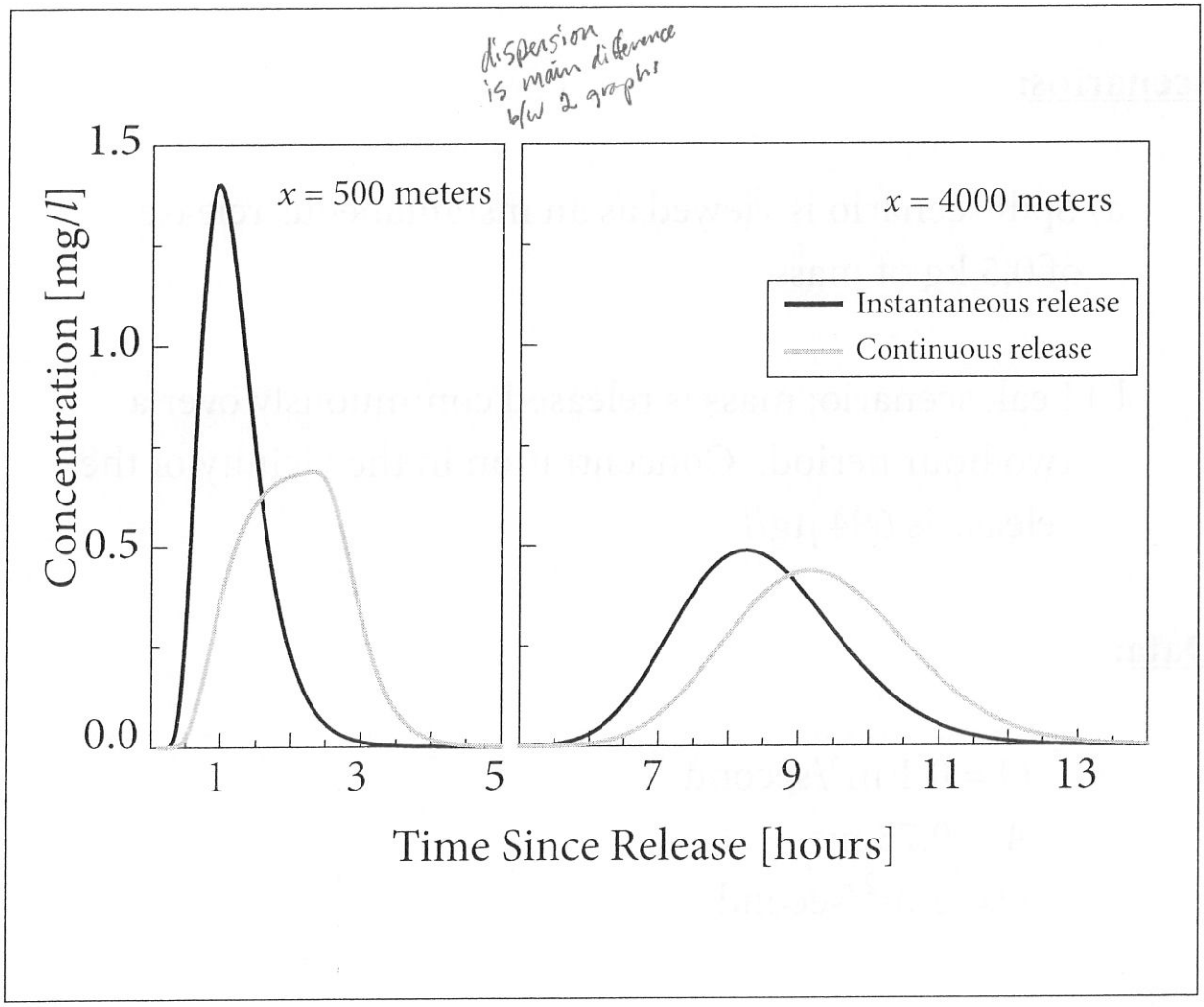
Data:

$Q = 0.1\ m^3/second$

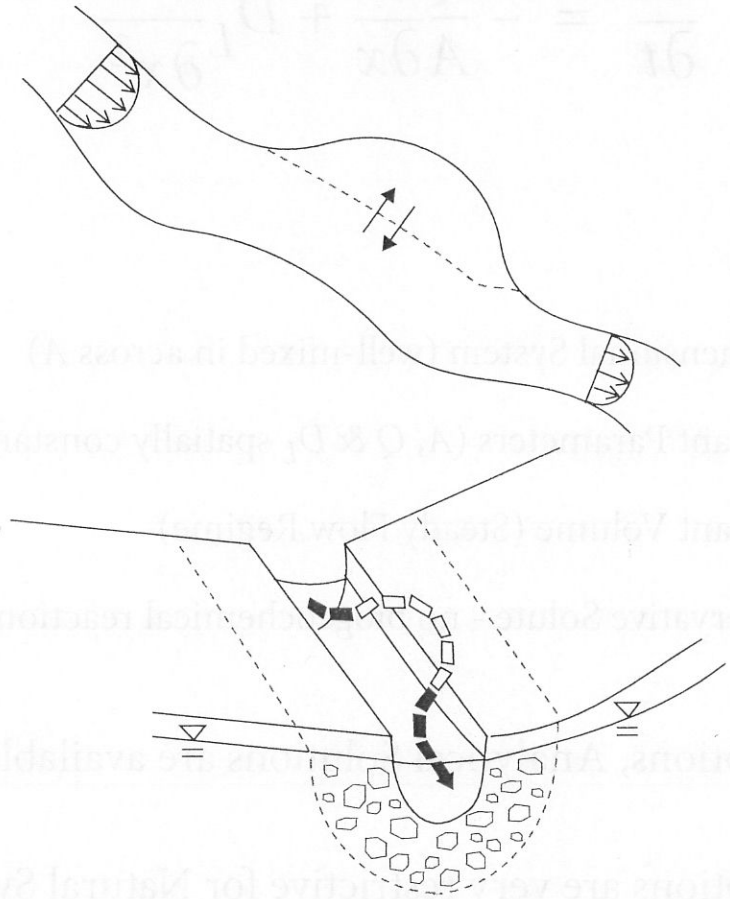
$A = 0.75\ m^2$

$D = 5\ m^2/second$

Stream Transport Processes: Advection-Dispersion Eqn
Analytical Solutions - Example



Stream Transport Processes II: Transient Storage



Stream Transport Processes: Constant Parameter, 1-Dimensional Advection-Dispersion Equation

$$\frac{\partial C}{\partial t} = -\frac{Q}{A}\frac{\partial C}{\partial x} + D_L\frac{\partial^2 C}{\partial x^2} \qquad (1)$$

Assumptions:

- 1-Dimensional System (well-mixed in across A)
- Constant Parameters (A, Q & D_L spatially constant)
- Constant Volume (Steady Flow Regime)
- Conservative Solute - no biogeochemical reactions

Given assumptions, Analytical Solutions are available

These assumptions are very restrictive for Natural Systems

Natural Systems

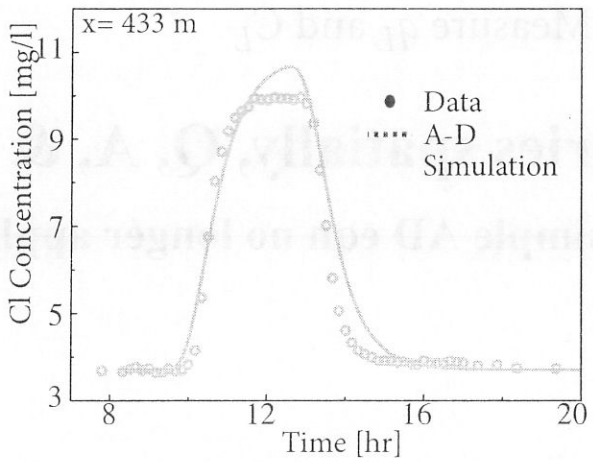
"I believe we are on an *irreversible* trend towards freedom and democracy -- but that could change" -- Dan Quayle

"Nothing is Constant"

Spatial Variability: Q, A, D_L vary in space

Unsteady Flow: Q & A vary in time

The Stream/Watershed Continuum
&
Non-Fickian Mixing



Spatially Variable Flow

Tributary Inflow

Point Sources of Solutes

Distributed Inflow

"Lateral Inflow"

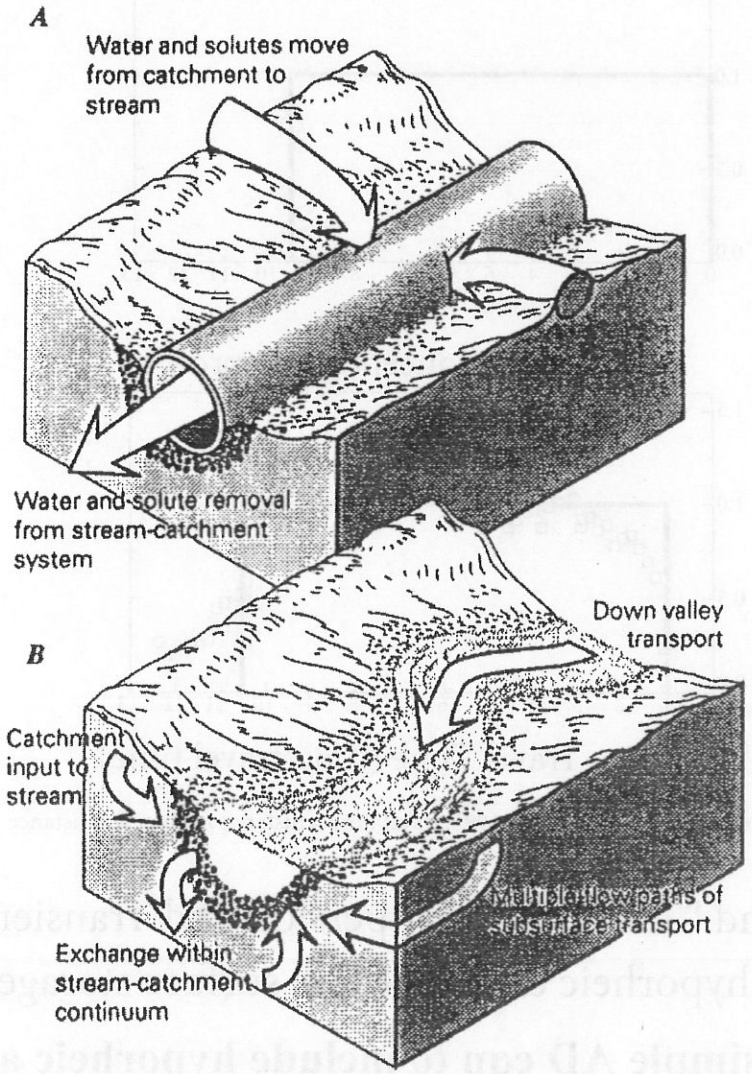
Sum of Groundwater inflow, Seeps/Springs, Overland flow

Distributed/Nonpoint Sources of Water & Solutes

Difficult to Measure q_L and C_L

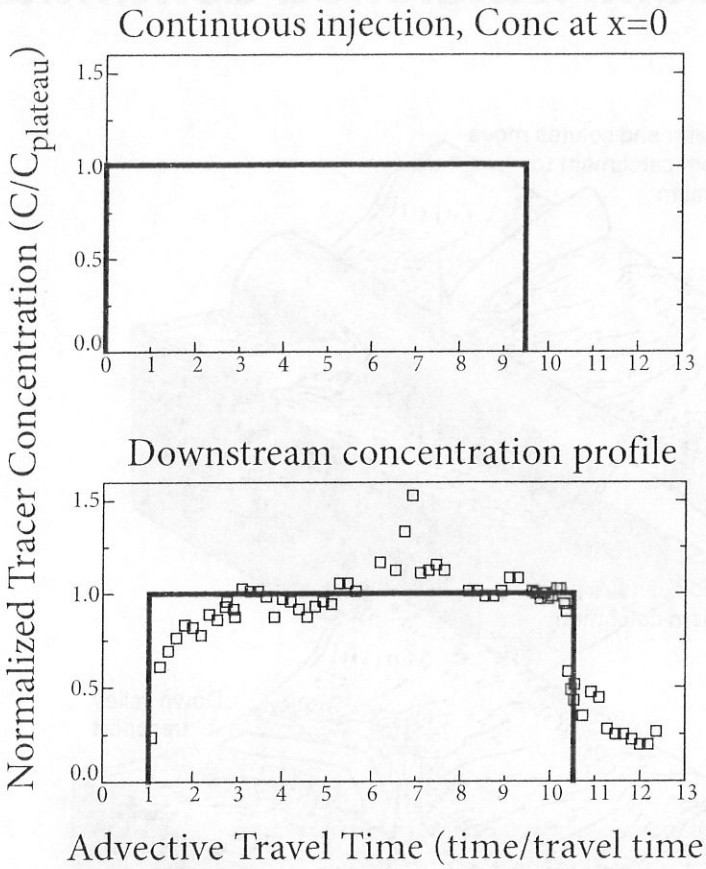
→ **If flow varies spatially, Q, A, & D vary spatially (simple AD eqn no longer applies).**

The Stream/Watershed Continuum



(Bencala, 1993)

Instream Mixing Processes



(Adapted from *Broshears et al.*, 1993; travel time = distance / velocity = distance / (Q/A))

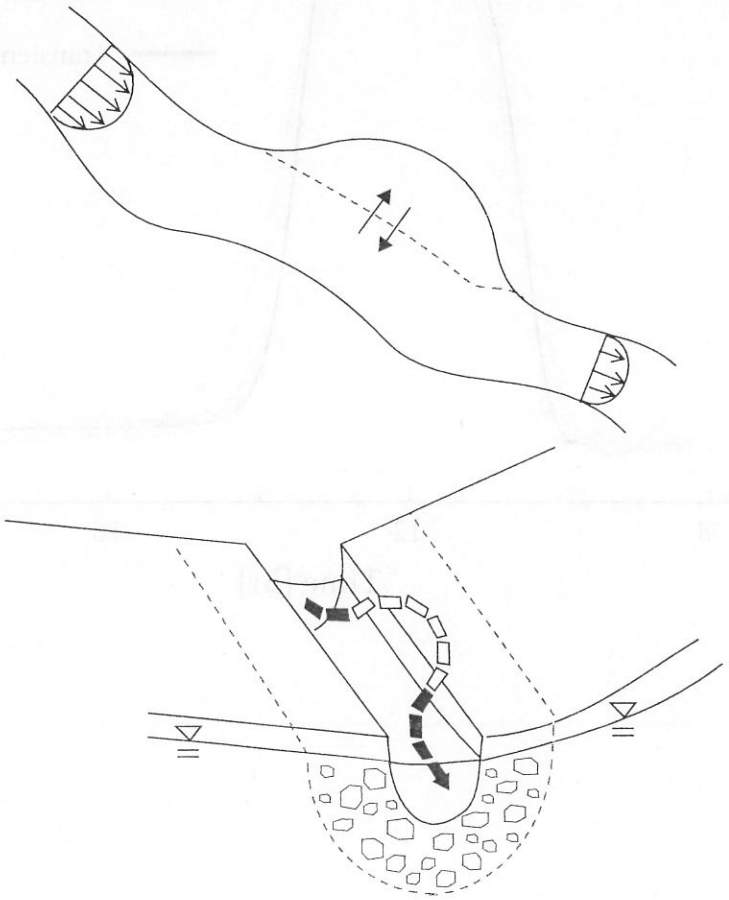
"Shoulder" and "Tail" due to Dispersion and Transient Storage (sheer stress, hyporheic exchange and surface storage)

→ **modify simple AD eqn to include hyporheic and surface storage**

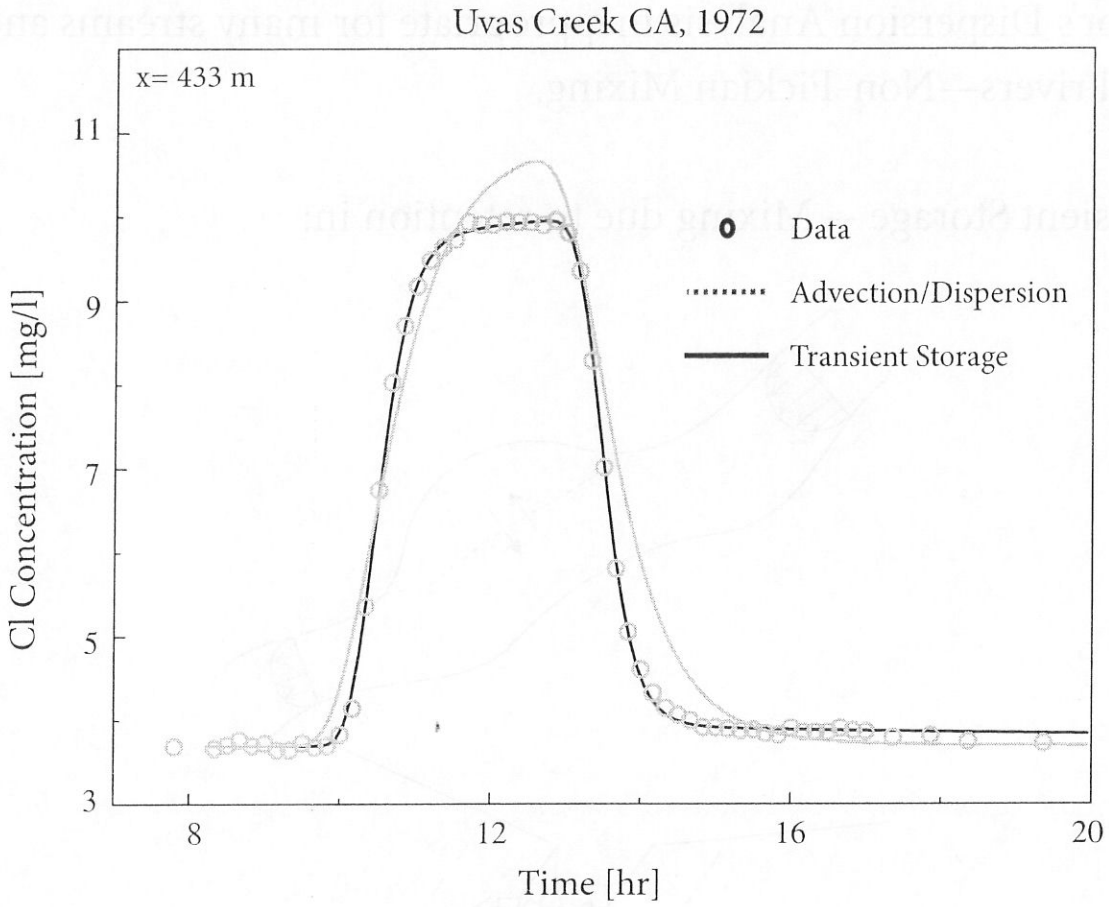
Transient Storage

Taylor's Dispersion Analysis inappropriate for many streams and small rivers—Non-Fickian Mixing.

Transient Storage -- Mixing due to retention in:



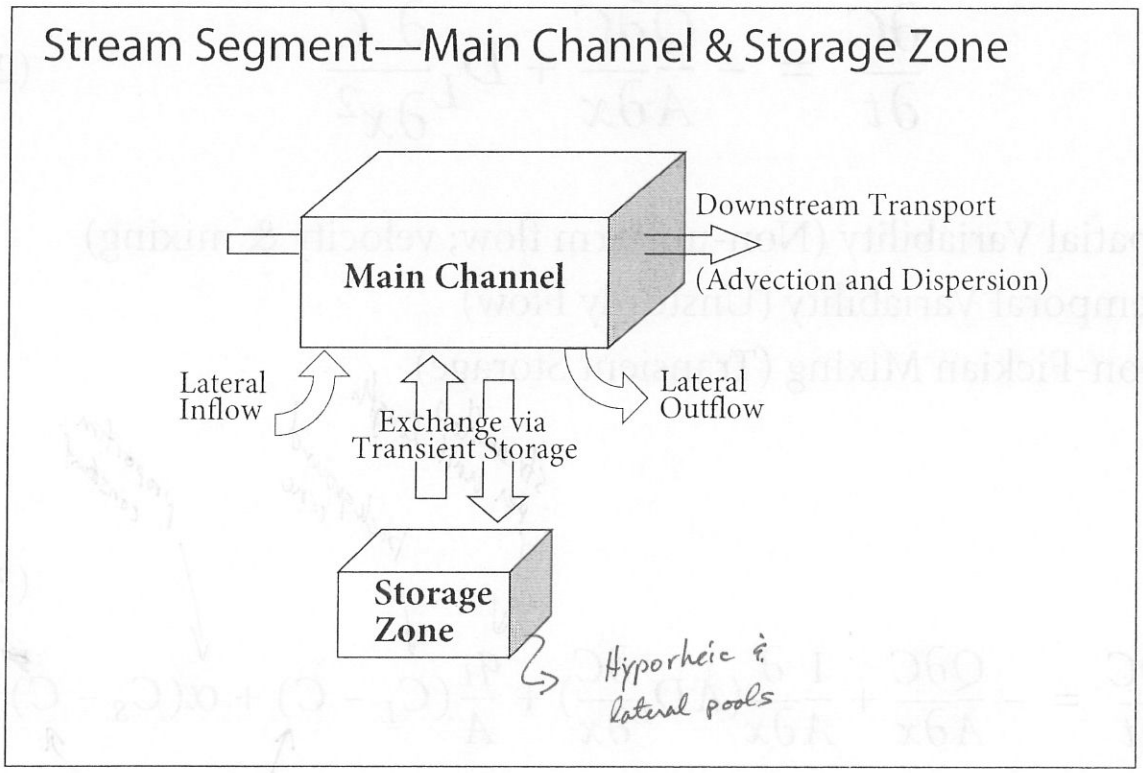
Transient Storage vs. Advection-Dispersion



OTIS-P

Conceptual Model: Main Channel and Storage Zone

Bencala & Walters (1983)



Governing Transport Equation: Main Channel

Constant Parameter, 1-Dimensional A-D Equation

$$\frac{\partial C}{\partial t} = -\frac{Q}{A}\frac{\partial C}{\partial x} + D_L\frac{\partial^2 C}{\partial x^2} \qquad (2)$$

+ Spatial Variability (Non-uniform flow; velocity & mixing)
+ Temporal Variability (Unsteady Flow)
+ Non-Fickian Mixing (Transient Storage)

=

streamflow per unit length

background conc

proportion constant

$$\frac{\partial C}{\partial t} = -\frac{Q}{A}\frac{\partial C}{\partial x} + \frac{1}{A}\frac{\partial}{\partial x}\left(AD_L\frac{\partial C}{\partial x}\right) + \frac{q_L}{A}(C_L - C) + \alpha(C_S - C) \qquad (3)$$

main channel conc

Advection **Dispersion** **Lateral Inflow** **Transient Storage**

$C_S < C$ as storage voids filling

$C_S = C$ at plateau

$C_S > C$ after plateau

Governing Transport Equations: Main Channel and Storage Zone

Main Channel

$$\frac{\partial C}{\partial t} = -\frac{Q}{A}\frac{\partial C}{\partial x} + \frac{1}{A}\frac{\partial}{\partial x}(AD_L\frac{\partial C}{\partial x}) + \frac{q_L}{A}(C_L - C) + \alpha(C_S - C)$$

Storage Zone

$$\frac{dC_S}{dt} = \alpha\frac{A}{A_S}(C - C_S)$$

Model Parameters

A - main channel cross-sectional area [L^2]

A_S - storage zone cross-sectional area [L^2]

D - dispersion coefficient [L^2/T]

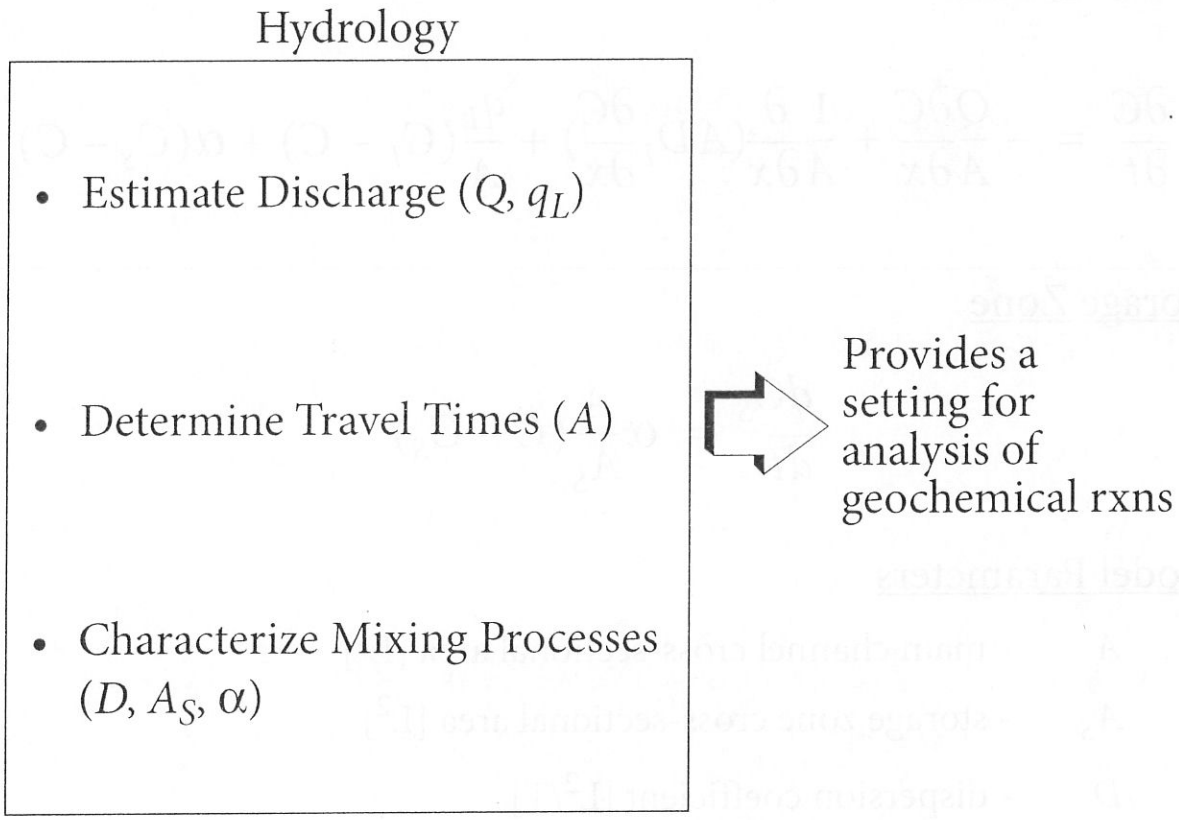
α - storage zone exchange coefficient [/T]

C_L - solute concentration in lateral inflow [mass/L^3]

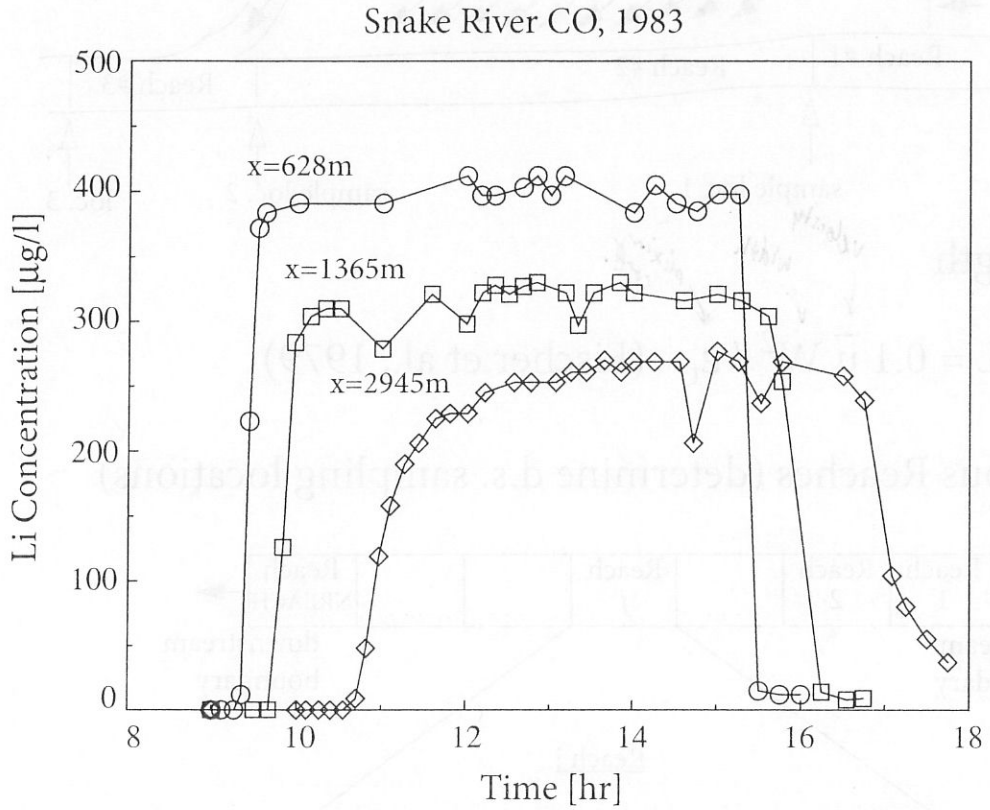
Q - volumetric flowrate; streamflow [L^3/T]

q_{LIN} - lateral inflow rate [L^3/T-L] — **Inflow per unit length**

Quantifying Transport Parameters: Tracer Methods



Downstream Profiles → Transport Parameters

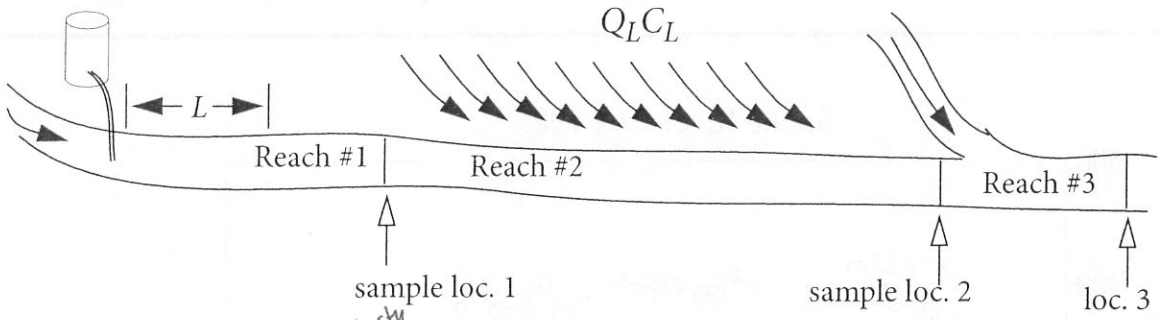


Magnitude --> Streamflow

Traveltime --> Velocity (Streamflow/Cross-Sectional Area)

Shape --> Mixing (Dispersion & Transient Storage)

Experimental Design

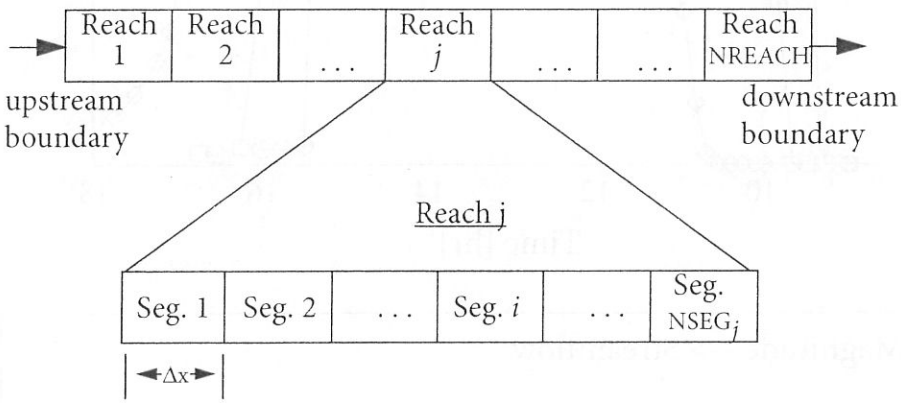


Mixing Length

velocity width mixing coeff.

$$L = 0.1\ \bar{u}\ W^2 / \varepsilon_t \quad \text{(Fischer et al., 1979)}$$

Homogeneous Reaches (determine d.s. sampling locations)



Conceptual System:

Reach - set of segments w/ uniform physical/chemical characteristics

Segment - main channel and storage zone control volumes in which mass is conserved (scale at which eqns & num. methods are applied)

Quantifying Transport Parameters

Model Inflow Concentration

(set equal to the background tracer concentration)

C_L - solute concentration in lateral inflow

Flow Parameters (estimated by Tracer-Dilution)

Q - volumetric flowrate; streamflow

q_{LIN} - lateral inflow rate (**inflow per unit length**)

$$q_{LIN} = \frac{Q_{bottom} - Q_{top}}{\text{reach length}}$$

Traveltime (velocity) and Mixing Parameters

(estimated via simulation using tracer [transport] data)

A - main channel cross-sectional area

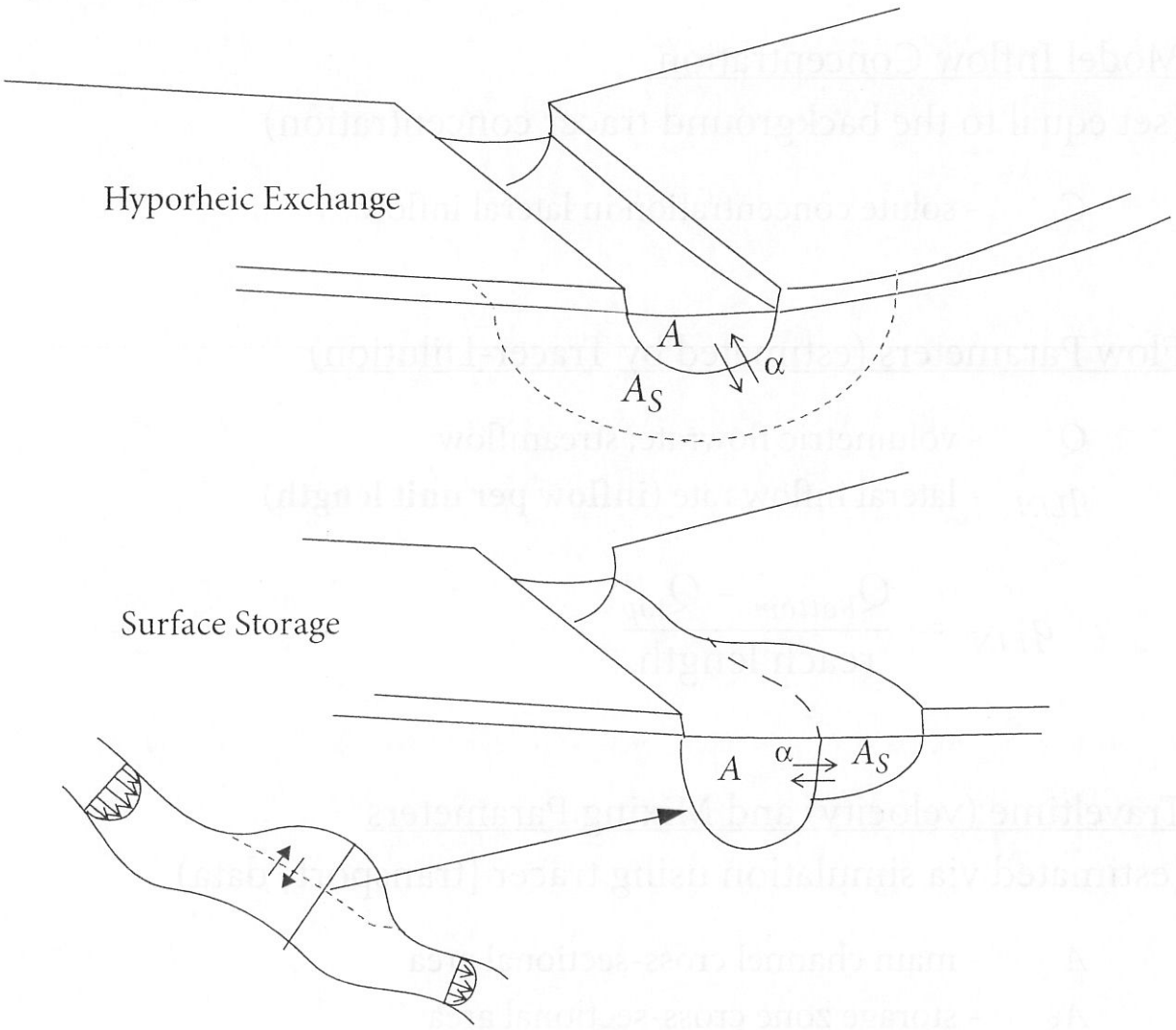
A_S - storage zone cross-sectional area

D - dispersion coefficient

α - storage zone exchange coefficient

Quantifying Transport Parameters—Transient Storage

Physical Interpretation of Model Parameters:



A_s and α: Determine Storage Zone Size and Solute Residence Time

storage area

transfer rate

Combined Tracer/Simulation Approach

Used to Estimate Parameters that are:

- Difficult to Measure in small systems (Q, q_L)
- Highly Variable (A)
- Abstract / Empirical (α, A_S, D)

$$\frac{\partial C}{\partial t} = -\frac{Q}{A}\frac{\partial C}{\partial x} + \frac{1}{A}\frac{\partial}{\partial x}\left(A D_L \frac{\partial C}{\partial x}\right) + \frac{q_L}{A}(C_L - C) + \alpha(C_S - C)$$

Main Channel Cross-Sectional Area (A)

Meaning

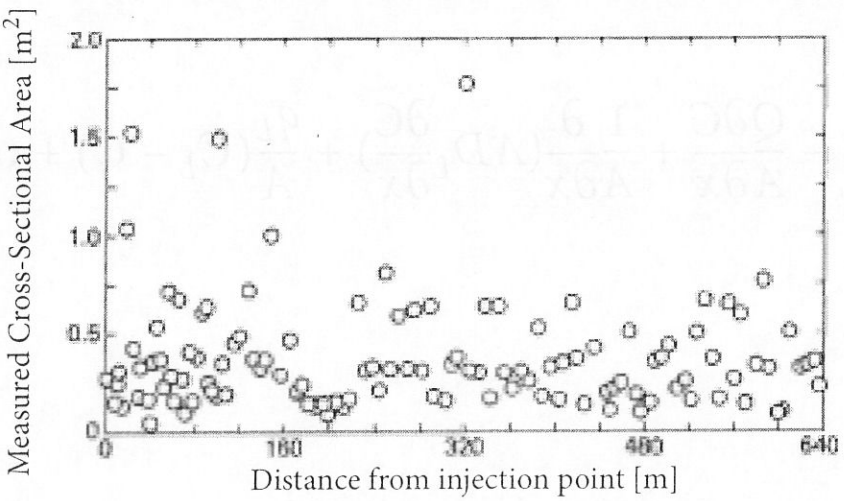
Average, over the entire reach, of the cross-sectional area (width x depth) for the main channel.

Average = mean of an infinite number of surveyed cross-sections

Tracer integrates the reach resulting in the average

For average to be representative, reach should be relatively homogeneous

Average Cross Sectional Area ≠ Measured Cross-Section

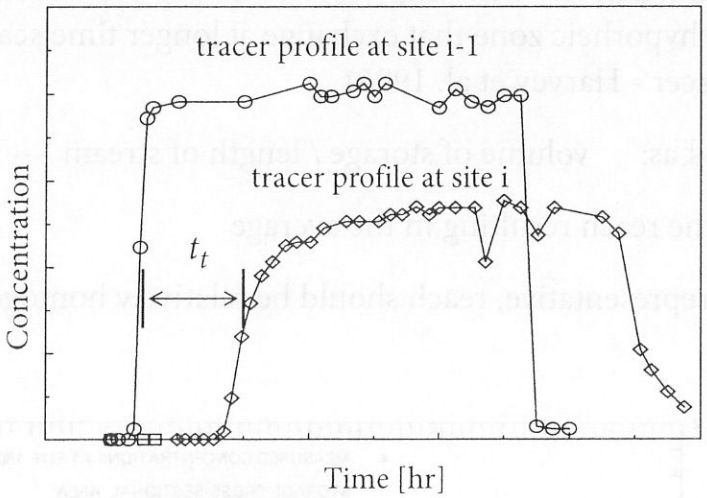


Measured channel cross-sectional area in Uvas Creek (Bencala & Walters, 1983)

Modeled cs area almost always smaller than measured cs area b/c modeled area only incorporates flowing water

Main Channel Cross-Sectional Area (A)

Initial Estimates of Main Channel Cross-Sectional Area—Tracer Traveltime



$$A = \frac{flow}{velocity} = \frac{(Q_i + Q_{i-1})/2}{(x_i - x_{i-1})/t_t} \quad (4)$$

where:

t_t — estimate of reach travel time, based on time at which C = 1/2 $C_{plateau}$

Storage Zone Cross-Sectional Area (A_S)

Meaning

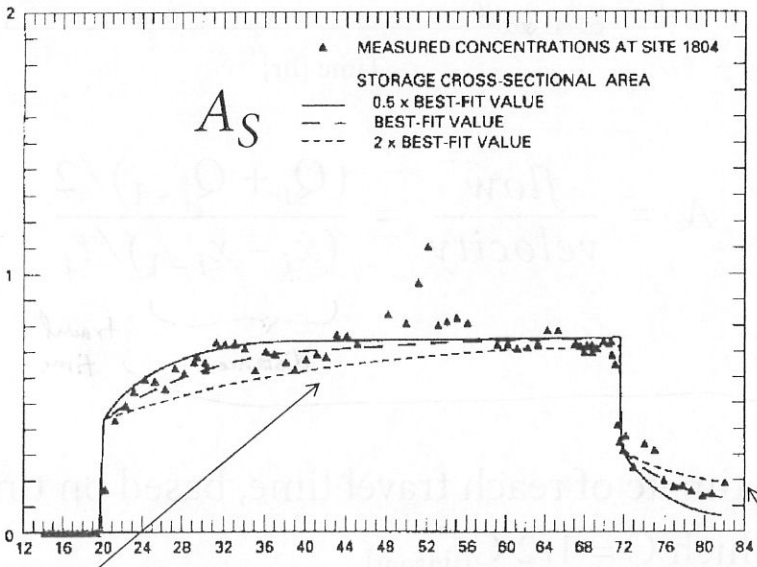
Average cross-sectional area of the storage zone—consisting of surface pools and hyporheic zones **that actively exchange tracer** with the main channel. (There may be portions of the hyporheic zone that exchange at longer time scales than can be detected by the tracer - Harvey et al. 1996)

Can also be viewed as: volume of storage / length of stream — $L^3/L = L^2$

Tracer integrates the reach resulting in the average

For average to be representative, reach should be relatively homogeneous

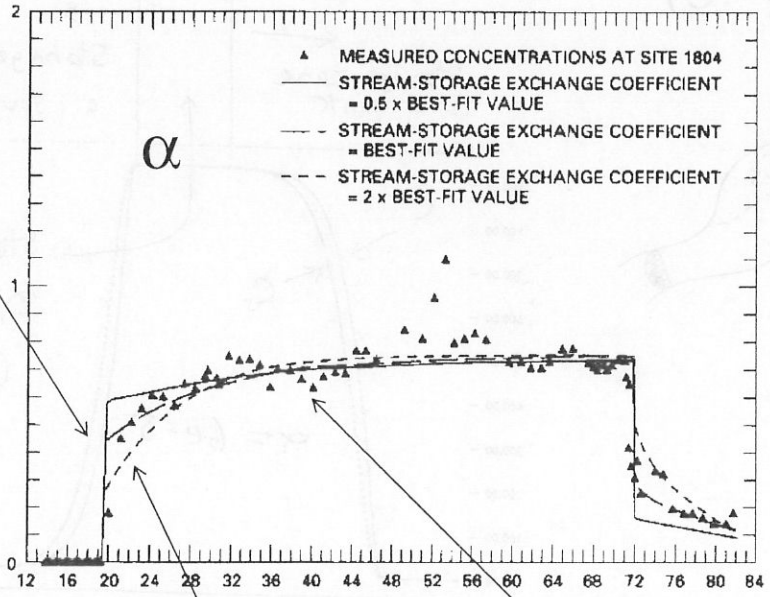
from *Broshears et al.* 1993



Larger A_S = larger volume to fill with tracer = longer time to plateau
= larger volume to empty = longer time to return to background

Storage Zone Exchange Coefficient (α)

α controls when the "shoulder falls away from the vertical"



from *Broshears et al.* 1993

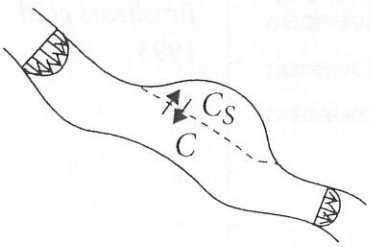
2 x best fit value (faster exchange)

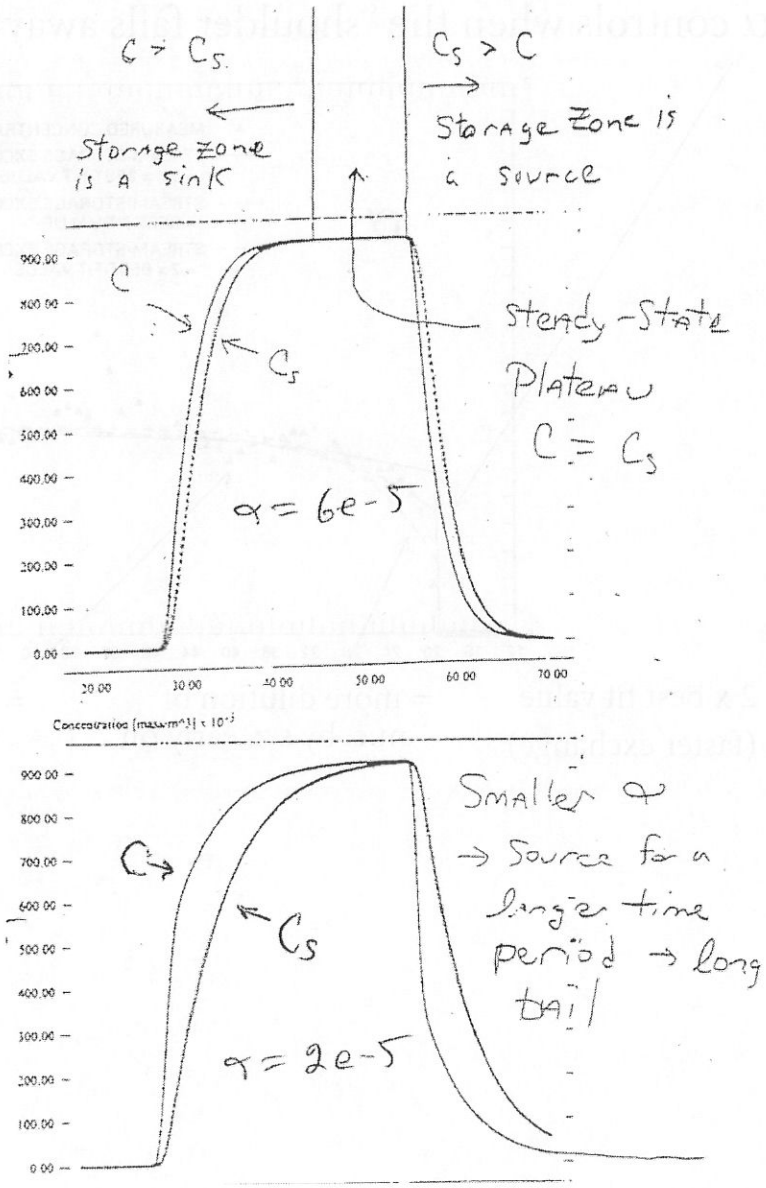
= more dilution of m.c. by s.z. early on

= earlier time to plateau

Storage Zone Exchange Coefficient (α)
Stream and Storage Zone Concentrations

$$\alpha(C_S - C)$$





Dispersion Coefficient (D)

In standard application of the A-D equation (no trans. storage)
estimated from mixing studies (dyes, natural tracers)
or
empirical eqns (D as a function of velocity, width, depth, slope)

In the presence of transient storage
estimated from the tracer data by simulation as with α and A_S

not comparable to D from standard application of the A-D equation -- some of the mixing is now attributed to storage. **Most (if not all) published empirical equations are therefore not applicable.**

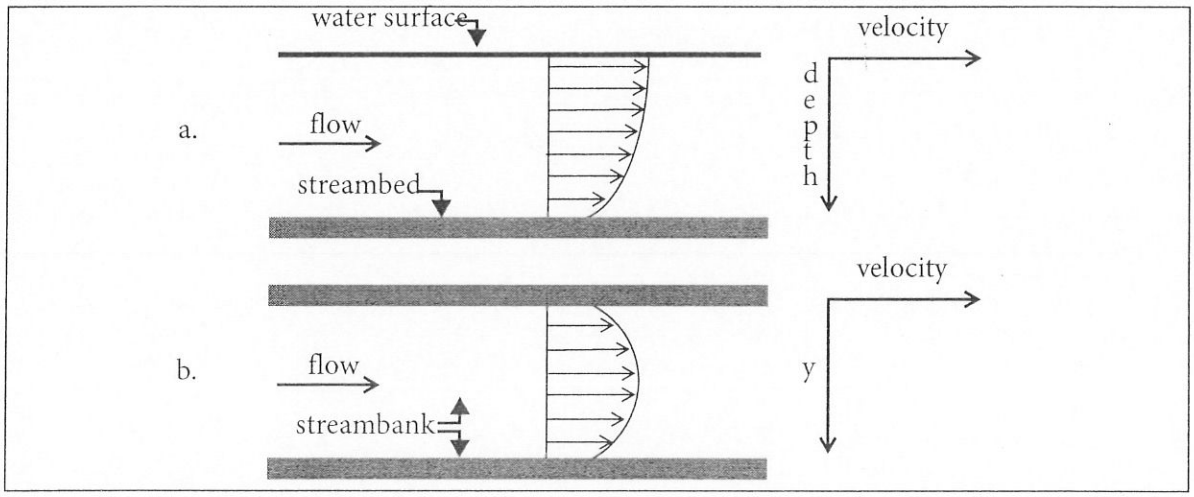
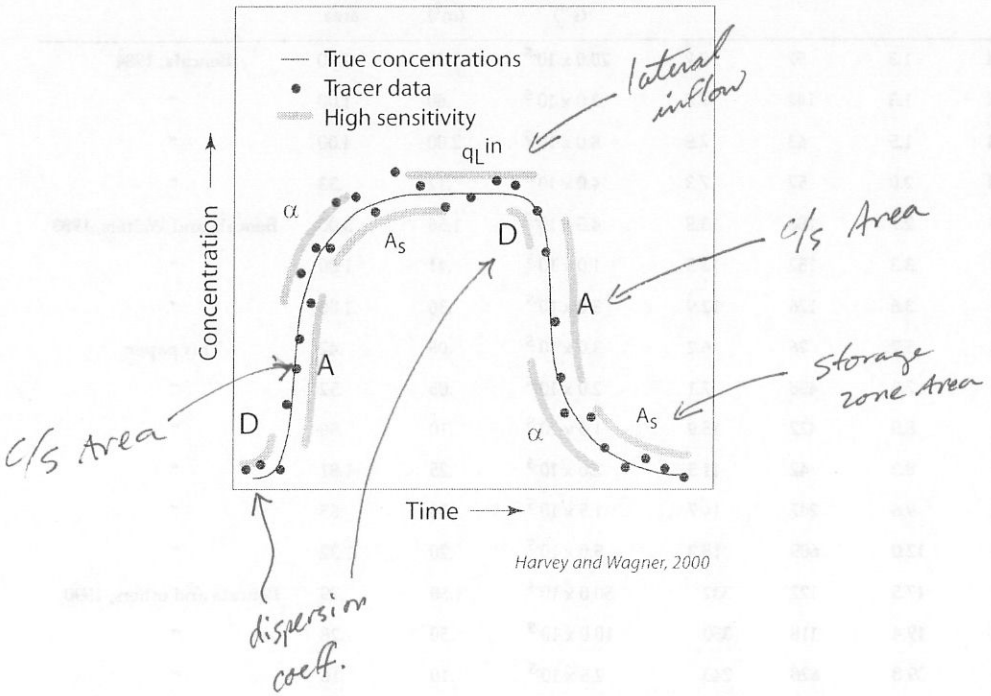


Table 2.01 (cont.), Summary of Dispersion Measurements in Streams

(EPA/600/3-87/015, 1987, Processes, Coefficients, and Models for Simulating Toxic Organics and Heavy Metals in Surface Waters)

Estimation of D, A_S, and α from Tracer Data: Model Sensitivity

(Harvey & Wagner, Book Chapter in "Streams and Groundwaters", 2000)



Transient Storage Parameters—Inter & Intra Stream Variability

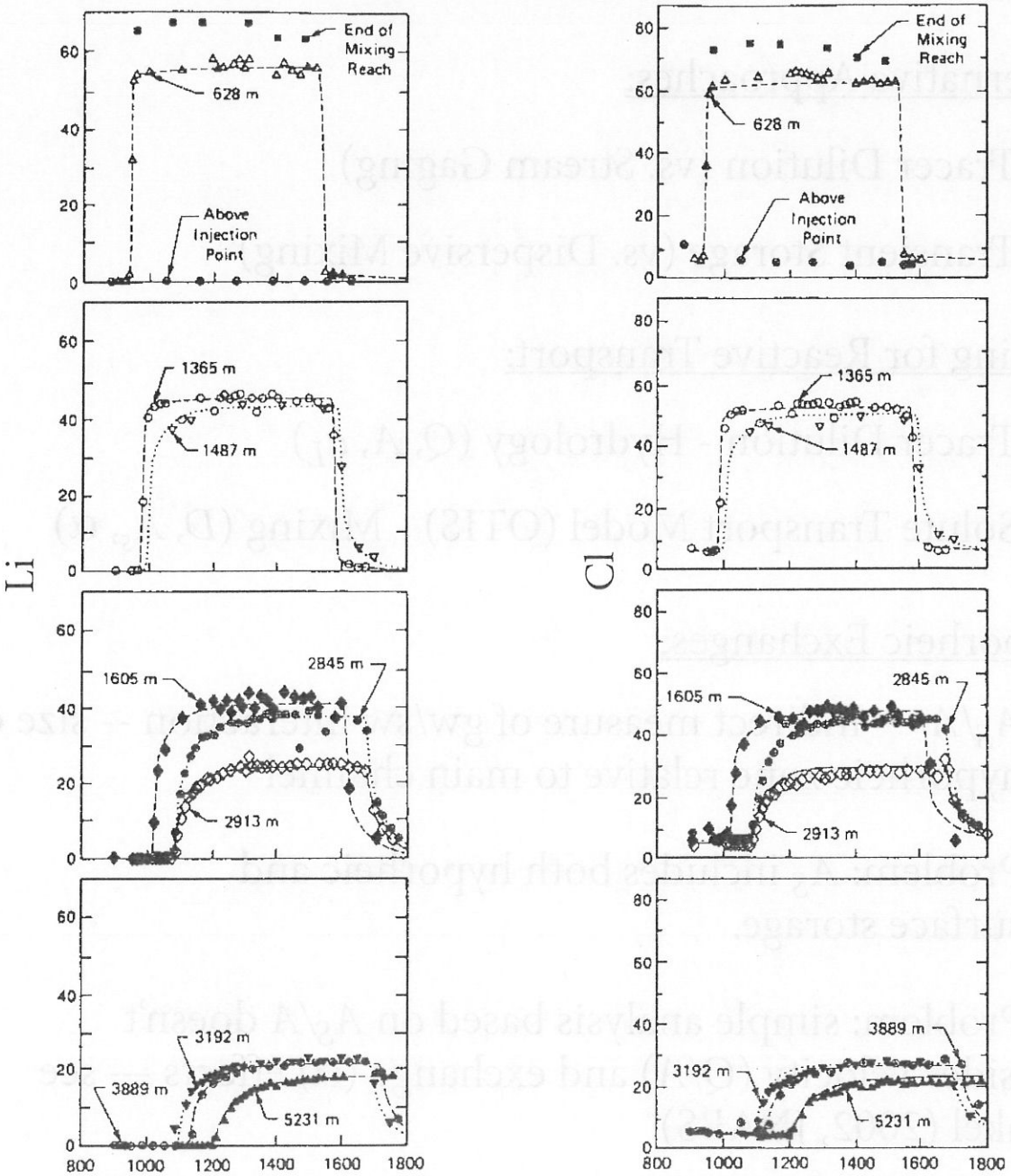
Broshears et al., 1993:

Stream	Water velocity (cm/s)	Reach length (m)	Average discharge (L/s)	Stream-storage exchange coefficient (s^{-1})	Storage cross-sectional area (m^2)	Ratio of storage area to stream area	Reference
LLM	1.3	57	7.7	20.0×10^{-5}	1.50	2.50	Bencala, 1984
LLM	1.3	142	7.8	2.0×10^{-5}	.60	1.00	"
LLM	1.5	63	7.5	8.0×10^{-5}	2.00	4.00	"
LLM	2.0	57	7.3	4.0×10^{-5}	.12	.33	"
UVS	2.7	186	13.8	4.5×10^{-5}	1.56	3.00	Bencala and Walters, 1983
UVS	3.3	152	13.5	1.0×10^{-5}	.41	1.00	"
UVS	3.6	176	12.9	3.0×10^{-5}	.36	1.00	"
SKG	5.2	26	6.2	3.0×10^{-5}	.05	.42	This paper
SKG	7.3	458	7.1	2.0×10^{-5}	.05	.52	"
SKG	8.0	422	15.9	1.5×10^{-5}	.10	.50	"
SKG	8.3	42	11.5	2.0×10^{-5}	.25	1.81	"
SKG	9.6	247	14.7	1.5×10^{-5}	.10	.65	"
SKG	12.0	609	18.3	5.0×10^{-5}	.20	1.32	"
SNK	17.5	122	332	50.0×10^{-5}	1.50	.79	Bencala and others, 1990
SNK	19.4	118	350	10.0×10^{-5}	.50	.28	"
SNK	39.8	628	243	2.5×10^{-5}	.10	.16	"
SNK	46.6	1,342	745	10.0×10^{-5}	.50	.31	"
SNK	53.4	737	294	5.0×10^{-5}	.02	.04	"
SNK	53.8	1,240	376	20.0×10^{-5}	.12	.17	"
SNK	55.4	697	670	5.0×10^{-5}	.05	.04	"
SNK	58.4	279	631	5.0×10^{-5}	.05	.05	"
SNK	62.5	68	500	20.0×10^{-5}	.05	.06	"

For a more comprehensive list, see tables in Runkel (2002, JNABS)

53 data sets

Conservative Behavior?



Bencala et al., 1990 — Simulations based on Li-derived Parameters reproduce the general features of the Cl data — lithium and chloride are transported in a similar manner.

Summary: Physical Process Descriptions

Alternative Approaches:

- Tracer Dilution (vs. Stream Gaging)
- Transient Storage (vs. Dispersive Mixing)

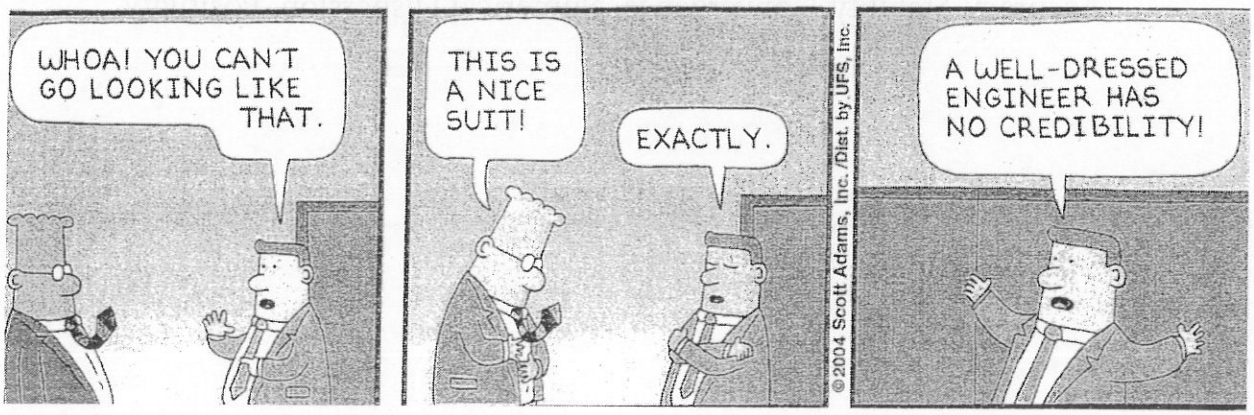
Setting for Reactive Transport:

- Tracer Dilution - Hydrology (Q, A, q_L)
- Solute Transport Model (OTIS) - Mixing (D, A_s, α)

Hyporheic Exchanges:

- A_S/A — indirect measure of gw/sw interaction -- size of hyporheic zone relative to main channel
- Problem: A_S includes both hyporheic and surface storage.
- Problem: simple analysis based on A_S/A doesn't consider velocity (Q/A) and exchange (α) effects — see Runkel (2002, JNABS)

Numerical Methods



Numerical Methods - Lecture Outline

I. Intro - Why this is really Easy

II. Solving a Simple Ordinary Differential Equation (ODE)

The Falling Parachutist, Euler's Method, The Fundamental Form

III. Solving our Favorite PDE - The Advection-Dispersion Eqn.

Finite Difference Methods: Explicit (e.g. Eulers), Implicit, Crank-Nicolson

IV. Important Terms / Concepts

Accuracy, Stability, Convergence, Numerical Dispersion, Positivity

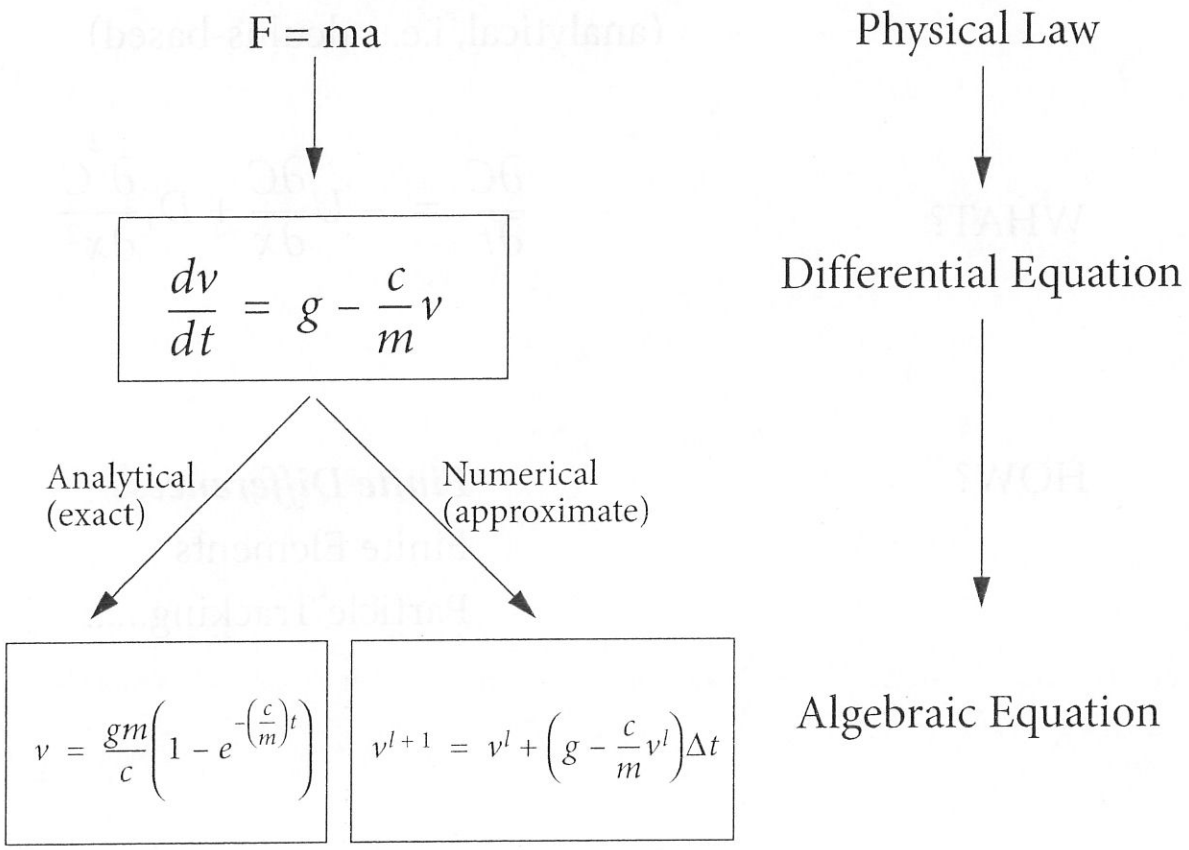
Numerical Methods

WHY? No analytical Solutions
(analytical, i.e. calculus-based)

WHAT? $$\frac{\partial C}{\partial t} = -U\frac{\partial C}{\partial x} + D_L\frac{\partial^2 C}{\partial x^2}$$

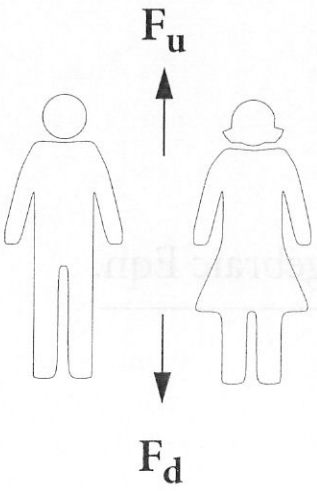
HOW? ***Finite Differences***
Finite Elements
Particle Tracking.....

Modeling:
Physical Law --> Algebraic Equation



(Chapra & Canale, 1988)

Newt's Falling Hydrologists



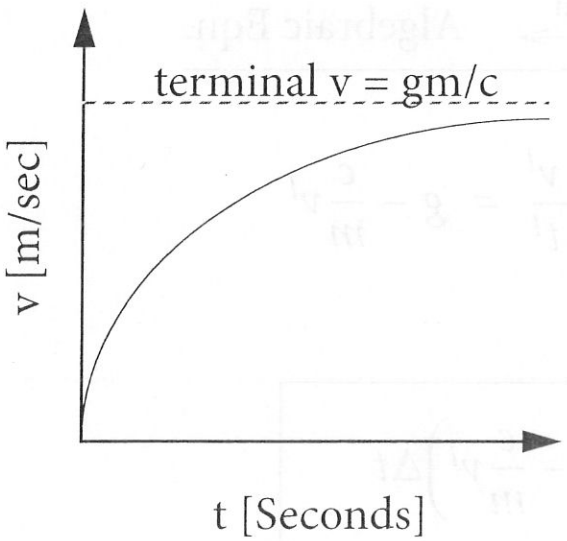
Predict the Velocity of a Falling Hydrologist

(After 'the falling parachutist', Chapra & Canale, 1988)

Fundamental Law —> Governing D.E.

$$ma = F$$

$$m\frac{dv}{dt} = mg - cv$$



$$\frac{dv}{dt} = g - \frac{c}{m}v$$

Analytical & Numerical Solutions

$$\frac{dv}{dt} = g - \frac{c}{m}v$$

Governing D.E. $\xrightarrow{\text{Analytical}}$ Algebraic Eqn.

$$v = \frac{gm}{c}\left(1 - e^{-\left(\frac{c}{m}\right)t}\right)$$

Governing D.E. $\xrightarrow{\text{Numerical}}$ Algebraic Eqn.

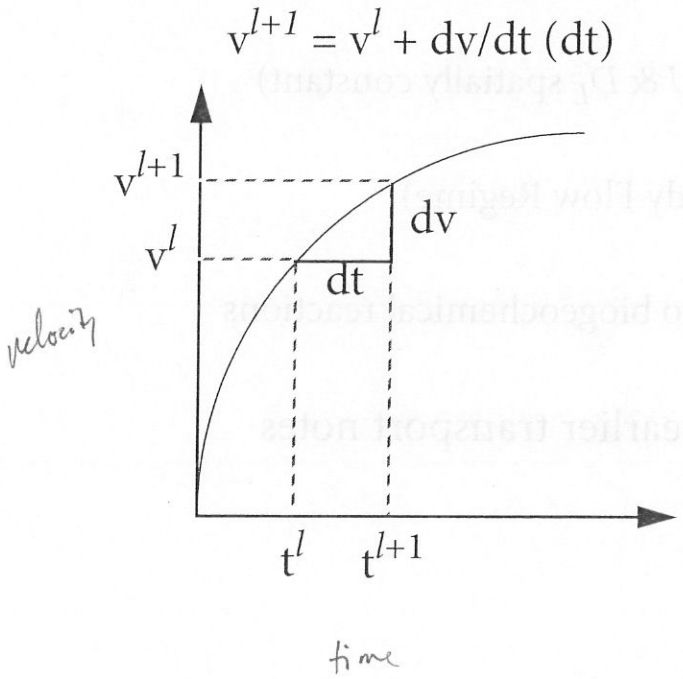
$$\frac{dv}{dt} = \frac{\Delta v}{\Delta t} = \frac{v^{l+1} - v^{l}}{t^{l+1} - t^{l}} = g - \frac{c}{m}v^{l}$$

$$v^{l+1} = v^{l} + \left(g - \frac{c}{m}v^{l}\right)\Delta t$$

Numerical Solution—Fundamental Form

$$v^{l+1} = v^l + \left(g - \frac{c}{m}v^l\right)\Delta t$$

$$NewValue = OldValue + Slope \times StepSize$$



Solution of PDEs:
The Advection-Dispersion Equation

$$\frac{\partial C}{\partial t} = -U\frac{\partial C}{\partial x} + D_L\frac{\partial^2 C}{\partial x^2} \quad (1)$$

Assumptions:

- 1-Dimensional System (well-mixed in across A)
- Constant Parameters (U & D_L spatially constant)
- Constant Volume (Steady Flow Regime)
- Conservative Solute - no biogeochemical reactions

Analytical Solutions: See earlier transport notes

Why use Numerical Methods? Natural Systems

"I believe we are on an *irreversible* trend towards freedom and democracy — but that could change" — Dan Quayle

"Nothing is Constant"

Spatial Variability: U (Q & A) & D_L vary in space

Unsteady Flow: Q & A vary in time

Additional Terms

Lateral Inflow, Transient Storage, Reactions

—> More complex transport equation cannot be solved analytically — it must be approximated numerically

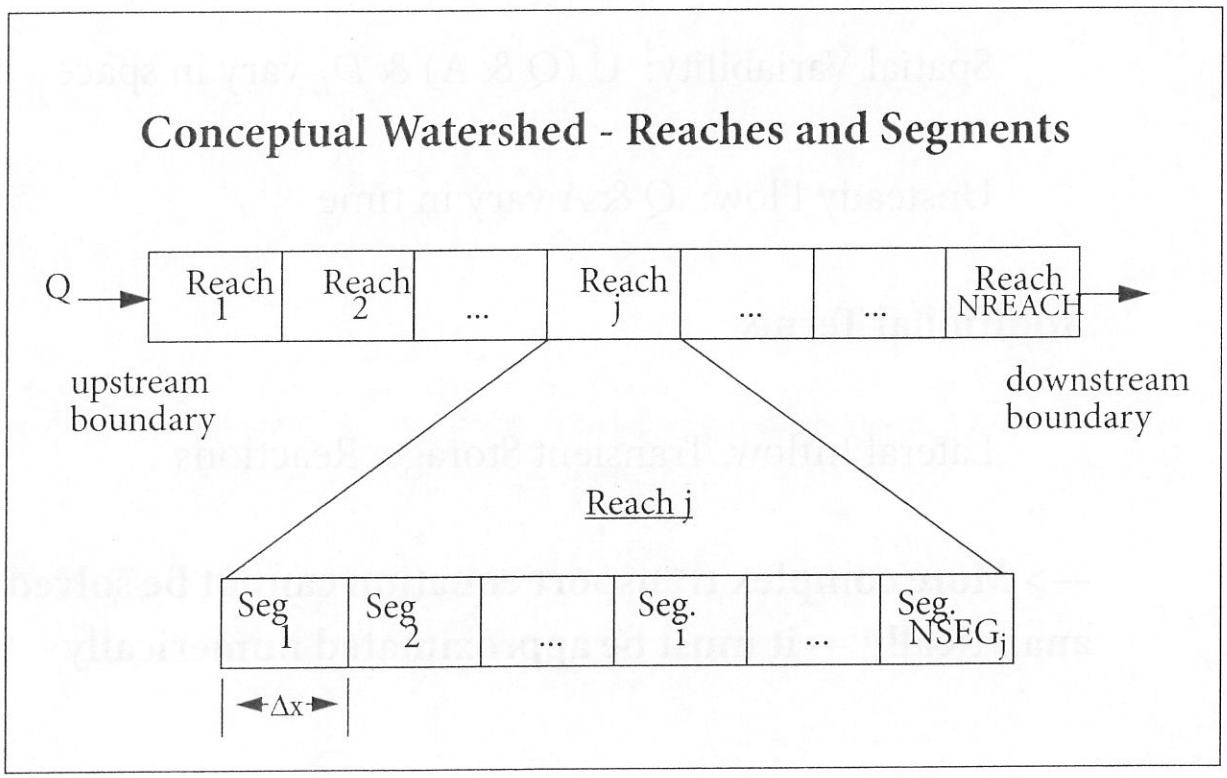
Characterizing Natural Variability
Reach-Specific Parameters

Reach - homogeneous length of stream

Reaches (Spatial Variability)

&

Segments (Mass Balance; Numerical Solution)



Numerical Solution — Adv-Disp Equation

$$\frac{\partial C}{\partial t} = -U\frac{\partial C}{\partial x} + D_L\frac{\partial^2 C}{\partial x^2}$$

$$\frac{C_i^{l+1} - C_i^l}{\Delta t} = Slope$$

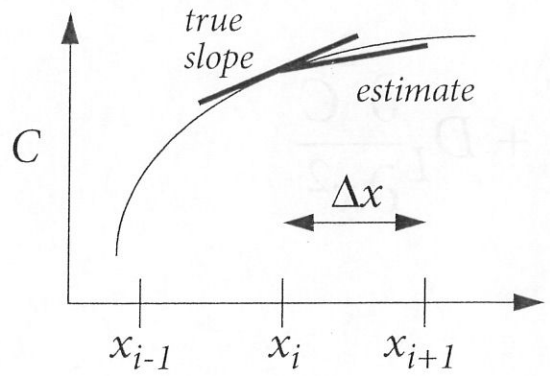
$$C_i^{l+1} = C_i^l + Slope(\Delta t)$$

Partial Differential Equation (PDE)

Temporal Derivative $\frac{\partial C}{\partial t}$

Spatial Derivatives $\frac{\partial C}{\partial x}$ $\frac{\partial^2 C}{\partial x^2}$

Finite Differences

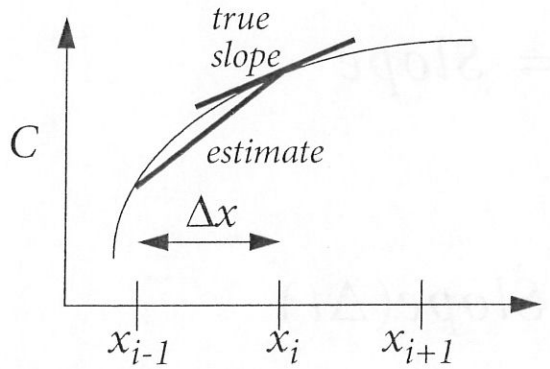


Forward Difference

$$\frac{dC}{dx} = \frac{C_{i+1} - C_i}{\Delta x}$$

Error ~ O(Δx)

(1st-order accurate)

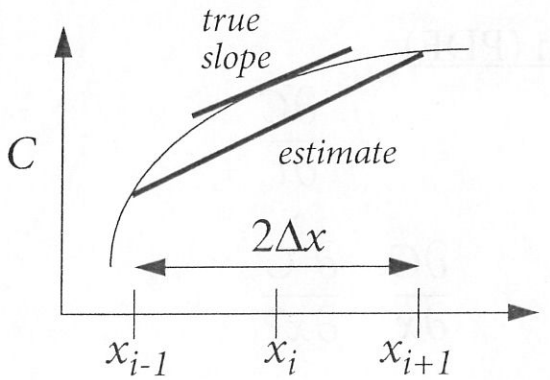


Backward Difference

$$\frac{dC}{dx} = \frac{C_i - C_{i-1}}{\Delta x}$$

Error ~ O(Δx)

(1st-order accurate)



Centered Difference (OTIS)

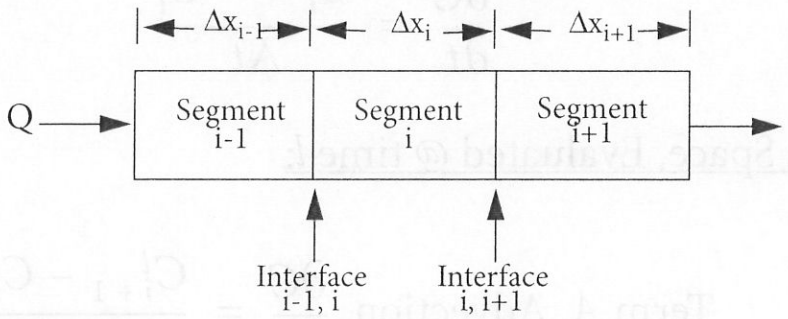
$$\frac{dC}{dx} = \frac{C_{i+1} - C_{i-1}}{2\Delta x}$$

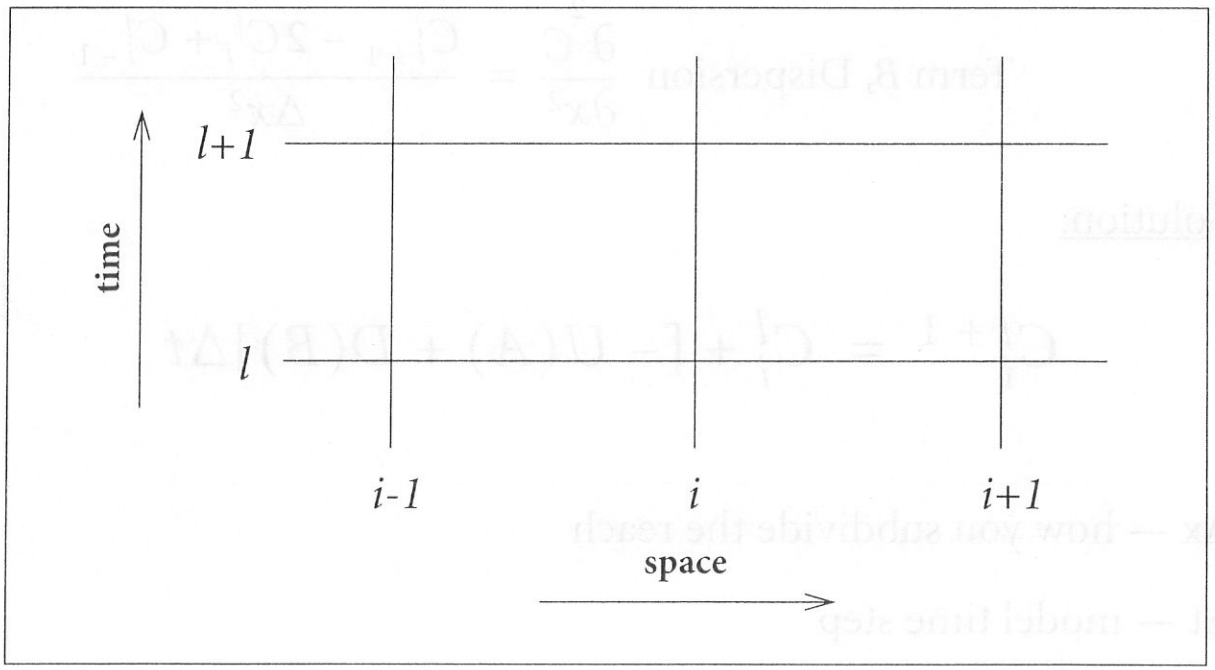
Error ~ $O(\Delta x^2)$

(2nd-order accurate)

Nomenclature

Segmentation Scheme





Explicit Solution - AD Equation

Forward in Time:

$$\frac{\partial C}{\partial t} = \frac{C_i^{l+1} - C_i^l}{\Delta t}$$

Centered in Space, Evaluated @ time *l*:

$$\text{Term } A\text{, Advection } \frac{\partial C}{\partial x} = \frac{C_{i+1}^l - C_{i-1}^l}{2\Delta x}$$

$$\text{Term } B\text{, Dispersion } \frac{\partial^2 C}{\partial x^2} = \frac{C_{i+1}^l - 2C_i^l + C_{i-1}^l}{\Delta x^2}$$

Solution:

$$C_i^{l+1} = C_i^l + [-U(A) + D(B)]\Delta t$$

Δx — how you subdivide the reach

Δt — model time step

Numerical Analysis - Criteria

- Accuracy:

  Error Analysis; e.g. Error ~ $O(\Delta t)$, $O(\Delta x^2)$

- Convergence:

  $\Delta t \rightarrow 0$ & $\Delta x \rightarrow 0$

  Approximate Solution —> True Solution

- Stability:

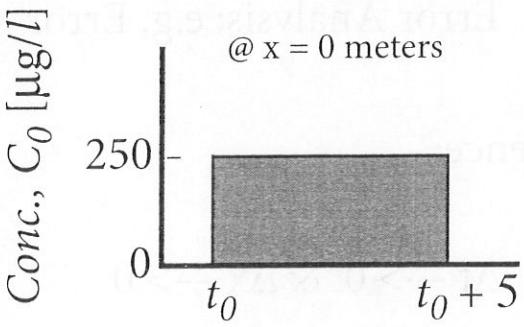
  Errors are Attenuated (not Amplified)

- Solution Positivity

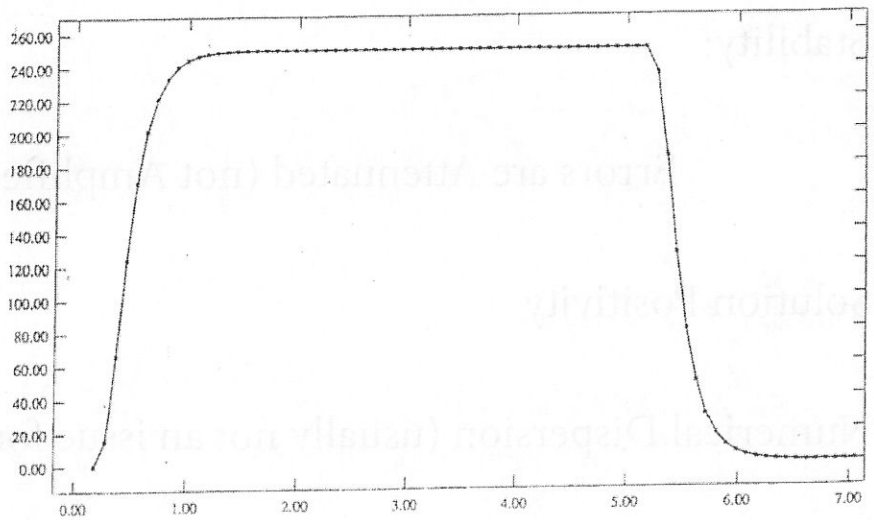
- Numerical Dispersion (usually not an issue for OTIS)

An Example

Leak scenario; mass is released continuously over a 5 hour period. Concentration in the vicinity of the release is 250 μg/*l*. Data: U = 3.5e-2 m^3/second, D = 0.2 m^2/sec, $\lambda = 0$.



Analytical Solution @ x = 50 Meters:

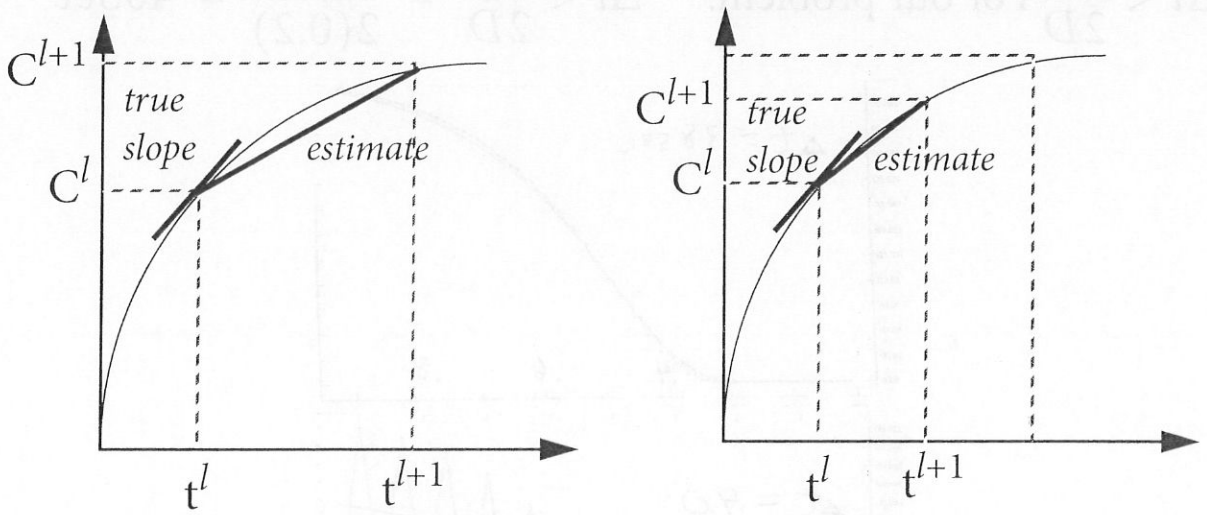


Accuracy and Convergence

Time Derivative - Forward Difference

Error ~ O(Δt), i.e. halving the time step —> halve the error

Convergence: As Δt—>0 & Δx—>0, Approx. Soln —> True Soln



Explicit vs. Analytic Solutions

Results @x=50 m, 32 min. after spill begins.

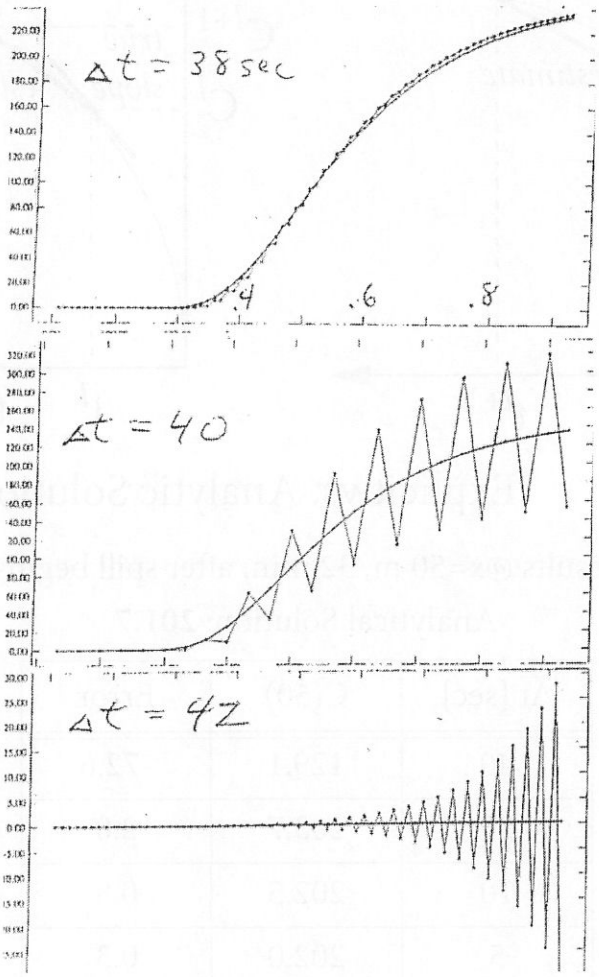
Analytical Solution: **201.7**

Δt [sec]	C(50)	Error
40	129.1	-72.6
20	203.7	2.0
10	202.5	0.8
5	202.0	0.3

Stability

Using the Explicit method and Central Finite Differences for the Advection term, the solution to the A-D equation will be stable, provided (Kent, 1960; Thomann & Mueller, 1987; WASP):

$\Delta t < \frac{\Delta x^2}{2D}$ For our problem: $\Delta t < \frac{\Delta x^2}{2D} = \frac{4^2}{2(0.2)} = 40 Sec$

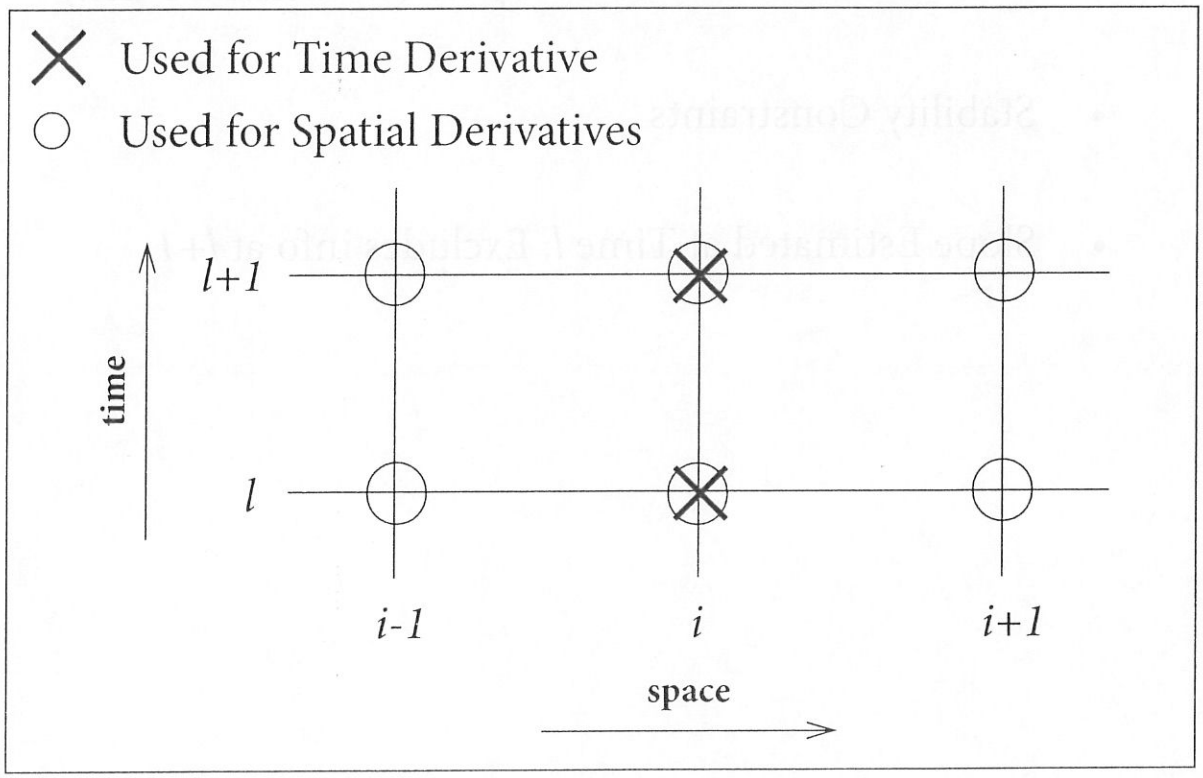


Summary - Simple Explicit Method

- Easy to Program
- Widely Used: WASP (U.S. EPA)
- First Order Accurate in Time
- Second Order Accurate in Space (w/ Central Diffs.)
- Stability Constraints
- Slope Estimated at Time *l*; Excludes info at *l+1*

Crank Nicolson (used in OTIS)

- Slope = (Slope@time *l* + Slope@time *l+1*) / 2
- 'Unconditionally Stable'
- Second-Order Accurate in Time



Method Comparison

Mass released continuously over a 5 hour period.
Concentration in the vicinity of the release is 250 μg/*l*.
Data: U = 3.5e-2 m^3/second, D = 0.2 m^2/sec.

vs. Analytic Solution

Results @ x=50 m, 32 min. after spill begins.

Analytic Solution: **201.7**

Δt [sec]	Explicit (Euler's)	Fully Implicit	Crank-Nicolson
640		158.2	152.2
320		174.5	187.6
160		186.0	194.9
80	-2.1e+12	193.2	198.2
40	129.1	197.2	199.8
20	203.7	199.3	200.6
10	202.5	200.3	201.0

Summary
Soln of the AD Equation

(You may now remove your Pocket Protectors)

Method Comparison

Method	Stability Constraint?	Temporal Accuracy	Numerical Dispersion
Explicit (Eulers)	Yes	First Order $E \sim O(\Delta t)$	Yes
Fully Implicit	No	First Order $E \sim O(\Delta t)$	Yes
Crank-Nicolson	No	Second Order $E \sim O(\Delta t^2)$	No[1]

1. There is no numerical dispersion provided central differences are used for the advection term (as in OTIS) and equal segment sizes are used (usually the case in OTIS).

Treatment of the Advection Term

Finite Difference	Accuracy	Numerical Dispersion	Solution Positivity
Backward	First Order $E \sim O(\Delta x)$	Increases Relative to central difference	Concentrations Always >0
Central (OTIS)	Second Order $E \sim O(\Delta x^2)$	Decreases Relative to backward difference	**Concentrations may go negative**

OTIS APPENDICES: Derivations, Finite Differences, C-N:

http://co.water.usgs.gov/otis/documentation/appendices/index.html

Notes on Use of OTIS— Input Files

Record Types & Fields

Each input file is described in terms of a set of '*record types*'. Within each of these record types, one or more '*fields*' is used to specify various input parameters. In general, record types refer to lines in the input file, while fields correspond to specific columns within each record.

Internal Comments

Comments are allowed anywhere in an input file, provided a # is placed in Column 1.

Units

- **Length Units [L].** Any unit of length may be used when specifying model input variables. The only requirement is that the length unit must be consistent for all model parameters and system configuration variables. The chosen length unit is denoted by an L in the input file descriptions that follow.
- **Concentration Units [CU].** The choice of concentration units is generally independent of the units used for the model parameters and system configuration variables. Consistent concentration units must be used for the specification of the boundary conditions (USBC, DSBOUND), the lateral inflow concentrations (CLATIN), and the sorption variables (RHO, KD, CSBACK). The chosen concentration unit is denoted by a CU in the input file descriptions that follow.
- **Time Units.** All model parameters that require the specification of a time unit (for example, flow rates and rate coefficients) are defined in terms of seconds. Note, however, that simulation control variables such as the simulation start time (TSTART) are specified in terms of hours

Integers and Floating Point Numbers

IF an input is an integer (I), don't use a decimal point

IF an input is double precision (D), use a decimal point. See Documentation....

Notes on Use of OTIS— Input Files & Other Info

If you have to work in Windows.....

(1) make sure you have a fixed font when editing in notepad, so all columns have the same width; (2) run from command prompt instead of double clicking, so you can see error msgs; (3) DAMIUL....

Beware the Tab — use spaces, not tabs

Always:

Work w/ OTIS before working w/ OTIS-P (debug your input files, get simulation "in the ballpark"

Check echo.out — verify parameters read in correctly; verify data read correctly; check Qs, etc.

Example OTIS Application
Steps 1 & 2: Initial Characterization of Transport w/o Modeling

How Fast Could TREX Run?

A new biomechanics model finds that T. rex lacked large enough leg muscles to run fast

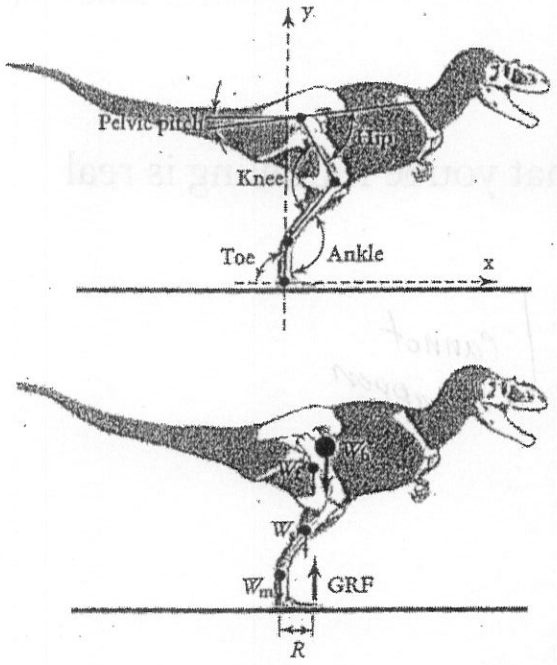


FIGURE 1. IN A FREE-BODY DIAGRAM for a running tyrannosaur, the angles of the leg joints (top) are critical parameters. In addition to the forces and torques produced by leg muscles, external forces (bottom) including the weights of the body (W_b), thigh (W_t), shank (W_s), and metatarsus (W_m), as well as the ground reaction force (GRF) that acts a distance R from the toe joint must be incorporated. (Adapted from ref. 6.)

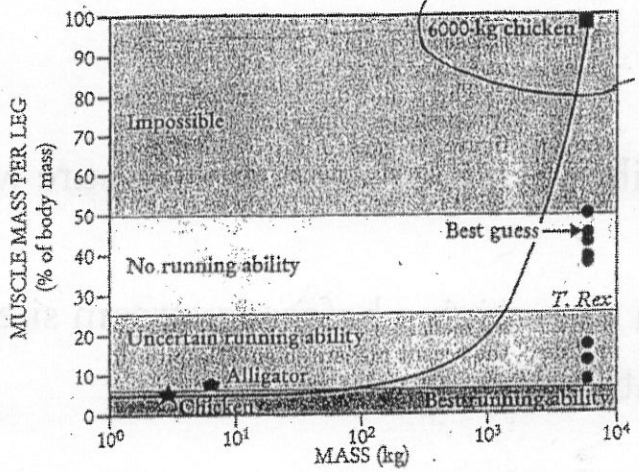


FIGURE 2. ESTIMATED EXTENSOR MUSCLE MASS per leg needed to run fast, as a fraction of total body mass. Larger animals need relatively more leg muscle to run. This dependence is illustrated by the solid line, calculated for a chicken scaled up to the size of *T. rex*. For several models of *T. rex* anatomy, the estimates (red) all require more leg muscle for running fast than the creature was likely to have had (orange dot). (Adapted from ref. 6.)

R. Fitzgerald, Physics Today, Apr 2002

Example OTIS Application: Steps 1 & 2, Initial Characterization of Transport

A Few Quick Words on Data Quality:

- Be a Skeptic
- Occam's razor
- Mission Impossible: QA your data to make sure what you're modeling is real

  e.g. downstream site arriving before upstream site; tracer concentration going up w/ distance | cannot happen

OTIS: One-Dimensional Transport with Inflow and Storage

- **Advection-Dispersion**
- **Transient Storage: *Bencala & Walters, WRR* 19(3), 1983**
- **Time-variable** and Steady-State Analysis of Solute Transport
- **Conservative Transport: Analysis of Tracer Data**
- Nonconservative Transport: First-Order Decay and Sorption
- Unsteady Flow Regimes [e.g. *Runkel et al., JNABS* 1998]
- Efficient Numerical Solution [*Runkel & Chapra, WRR* 1993]
- **Automated Parameter Estimation (OTIS-P)**
- Public Domain Software, http://co.water.usgs.gov/otis

Runkel, R.L., 1998, One dimensional transport with inflow and storage (OTIS): A solute transport model for streams and rivers: U.S. Geological Survey Water-Resources Investigation Report 98-4018. 73 p.

Basic use of OTIS for analyses of Transient Storage

- Continuous, Constant Rate Tracer Injection
- Conservative Transport
- Steady Flow
- Single Solute (tracer)
- Time Variable Simulation
- Manual Trial-and-Error Parameter Estimation
- Analysis of Tracer Data at One Location Only—Single Reach

Use OTIS first!

Then use OTIS-P

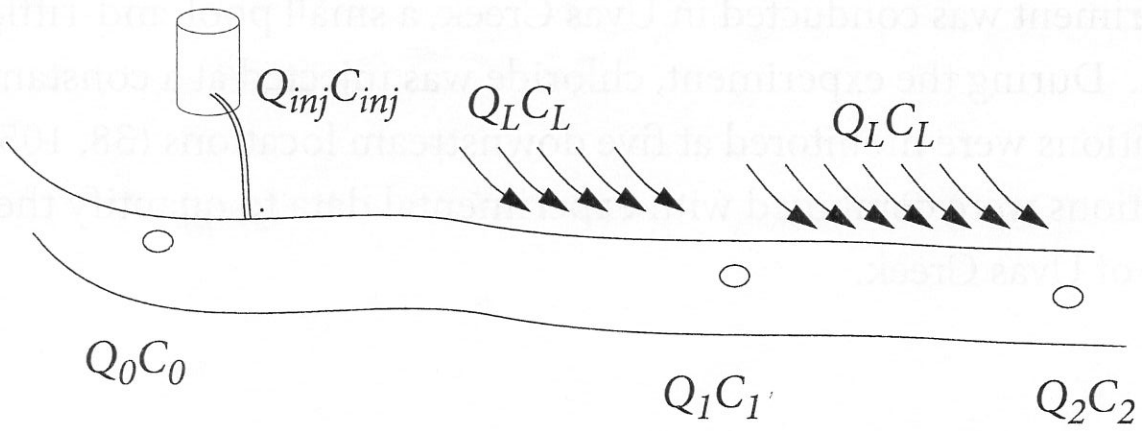
Example Application: Uvas Creek California

Bencala and Walters (1983) describe an instream experiment in which chloride is used as a conservative tracer. The experiment was conducted in Uvas Creek, a small pool-and-riffle stream in northern California. During the experiment, chloride was injected at a constant rate for 3 hours and concentrations were monitored at five downstream locations (38, 105, 281, 433, and 619 m). Simulations were combined with experimental data to quantify the hydrodynamic characteristics of Uvas Creek.

- 3 Hour Tracer Injection
- Conservative Transport of Chloride subject to Steady Flow
- Analysis of Tracer Data at 105 meters
- Manual Trial-and-Error Parameter Estimation (OTIS)
- Automated Parameter Estimation (OTIS-P)

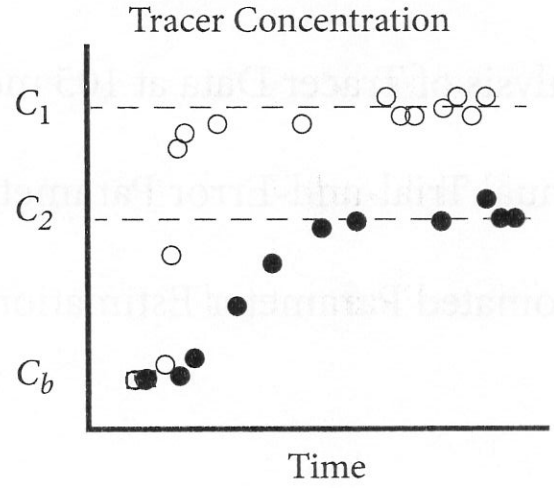
Step 1 - Estimate Streamflow

Q by Tracer-Dilution - Constant Rate Injection



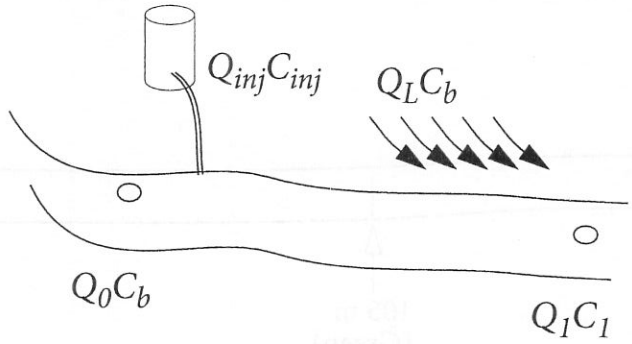
Assume:

* UNIFORM BACKGROUND concentration of tracer in stream and inflows ($C_0 = C_L = C_b$).
  NOTE: not always the case — e.g. road salt
* $Q_{inj} << Q_0$
* $Q_{inj}C_{inj}$ = constant in time
* Steady Flow



Step 1 - Estimate Streamflow

Eqns for Q by Tracer-Dilution—**subject to assumptions on previous page**



Flow Balance (Flow in = Flow out)

$$Q_0 + Q_L = Q_1 \tag{1}$$

Mass Balance (Mass in = Mass out)

$$Q_0C_b + Q_LC_b + Q_{inj}C_{inj} = Q_1C_1 \tag{2}$$

Combine Flow and Mass Balance

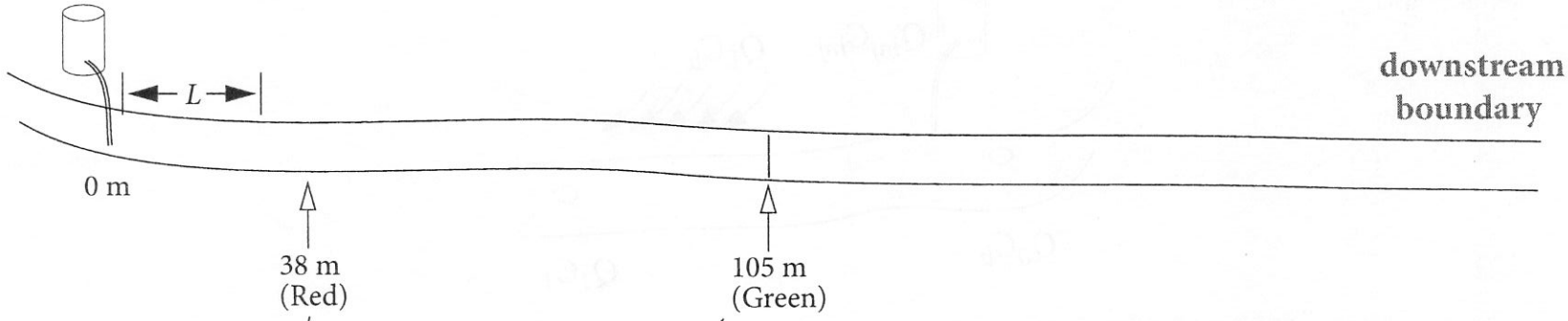
Use flow balance to eliminate Q_0 and Q_L from mass balance and solve for Q_1:

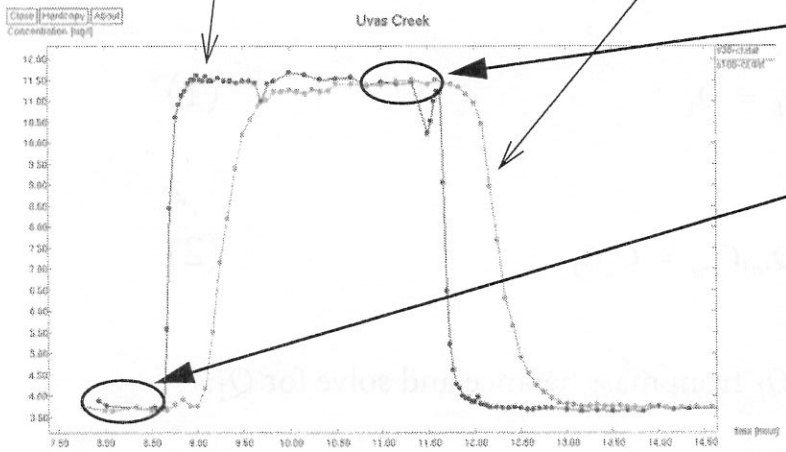
$$Q_1C_1 = Q_1C_b + Q_{inj}C_{inj}$$

$$Q_1 = \frac{Q_{inj}C_{inj}}{C_1 - C_b} \tag{3}$$

Step 1 - Estimate Streamflow

Uvas Creek—Single Reach OTIS Application





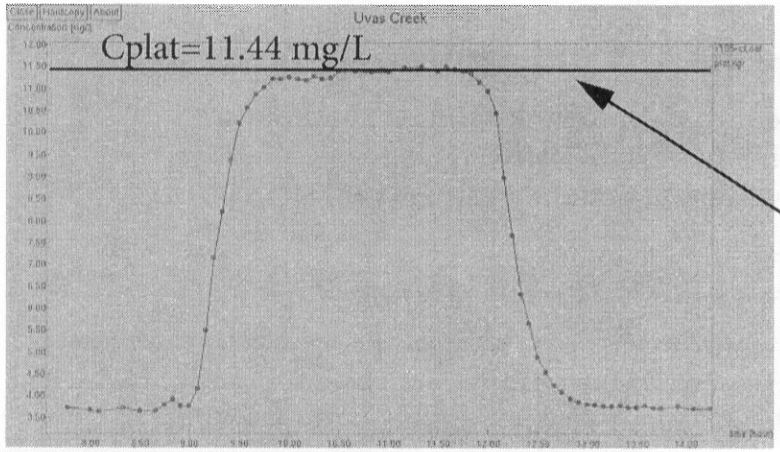
1) Chloride plateau at 38 and 105 m is the same
→ there's no inflow between inj & 105 m

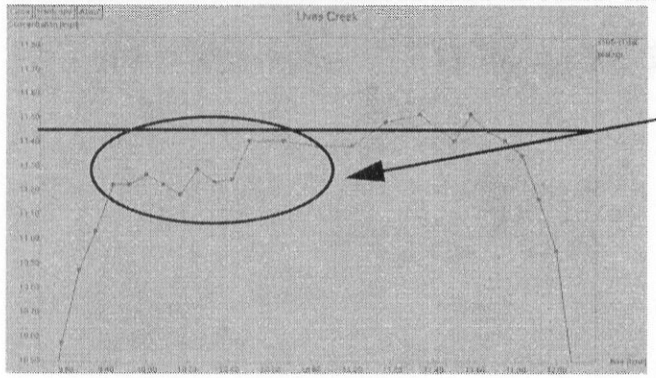
2) Background chloride = 3.7 mg/L

More on the downstream boundary condition later.....

Step 1 - Estimate Streamflow

Uvas Creek—Estimate Q based on plateau at 105 m





$$Q_1 = \frac{Q_{inj} C_{inj}}{C_1 - C_b}$$

Qinj = 50 mL / min
Cinj = 118.7 g Cl / L
Cb = 3.7 mg/L
C_1 = plateau concentration

1) define plateau—when does conc. stop rising?

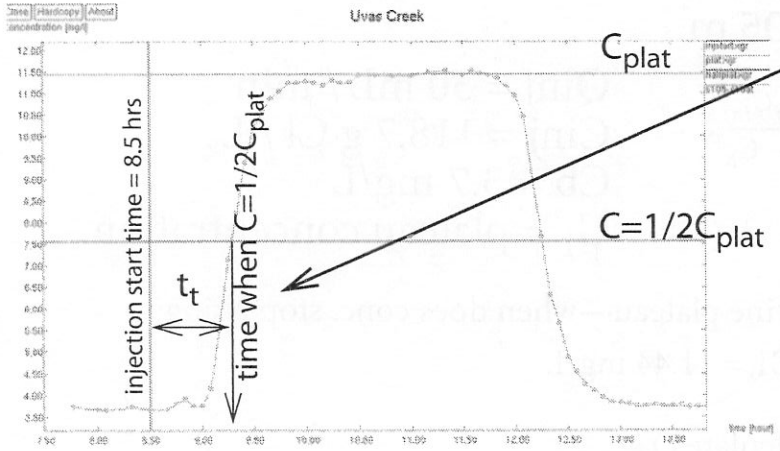
➔ C1 = 11.44 mg/L

2) calculate Q

➔ Q = 0.0128 m3 / s 12.8 L/s

3) definition of the plateau is crucial—if you place it too low, the 'shoulder' of the tracer arrival won't be recoginized & you'll miss the storage

Step 2—Obtain Initial Estimates of Traveltime, Velocity, & Main Channel Area



1) traveltime (t_t) = 9.29hrs-8.5hrs = 2826 s

2) velocity=distance/ttime=105 m/2826 s=0.037 m/s

Note: I made a mistake and used a distance of 150 m. For illustrative purposes, let's use my erroneous velocity estimate of 0.053 m/s

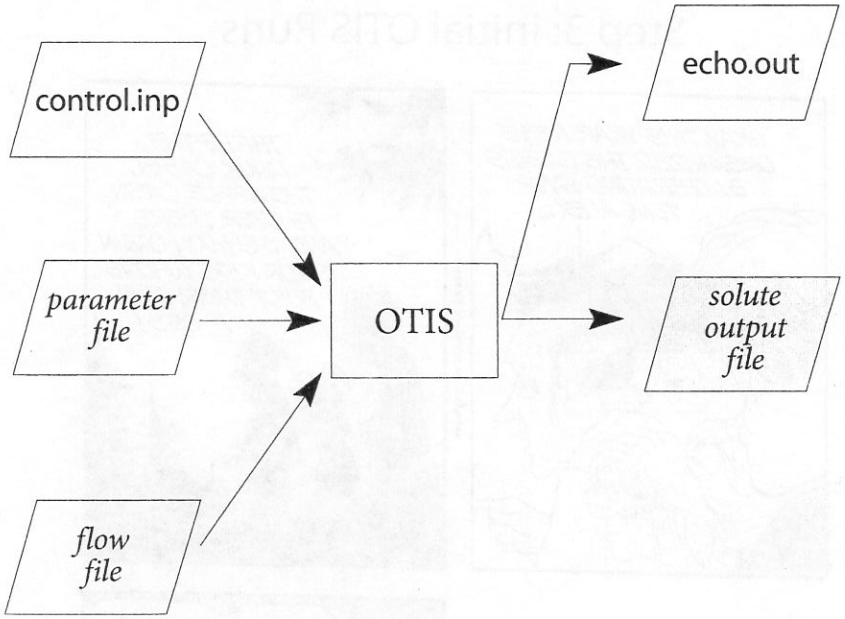
3) area = flow / velocity
= 0.0128 m^3/s / 0.053 m/s = 0.24 m^2

Note: for more complex cases, i.e. within reach flow increases and/or multiple reaches, see earlier lecture notes on main-channel cross-sectional area.

Finally, note that these are INITIAL ESTIMATES—We'll develop final estimates when the modeling steps are complete.

Example OTIS Application
Step 3: Initial OTIS Runs

Step 3—Initial OTIS Runs (NOT OTIS-P!)



Input Files

control.inp — used to specify the filenames of the remaining input and output files

parameter file — sets simulation options, boundary conditions, and model parameters that remain constant throughout the run.

flow file — contains model parameters related to flow and velocity (volumetric flow rate and main channel cross-sectional area).

Output Files

echo.out — contains an "echo" of the user-specified simulation options and model parameters. Also contains any error msgs generated during model execution.

solute output file — m.c. and s.z. conc at specified locations

Step 3—Initial OTIS Runs

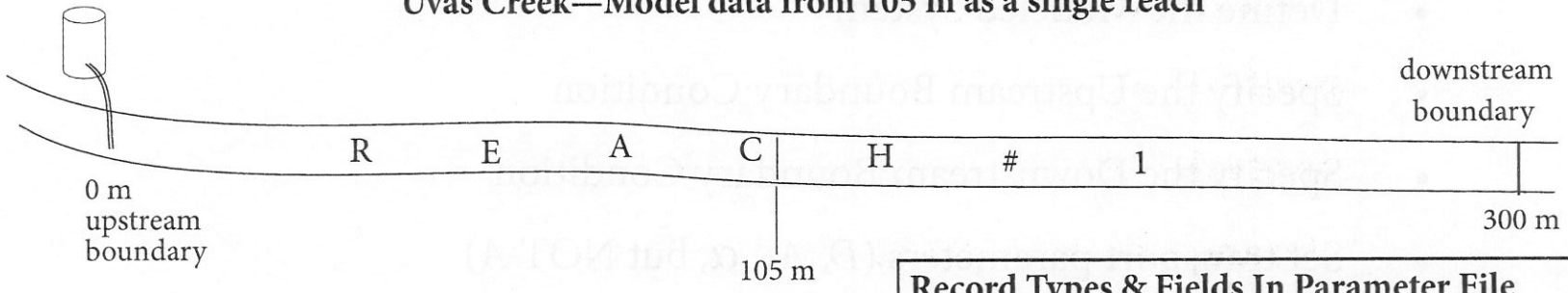
Parameter File:

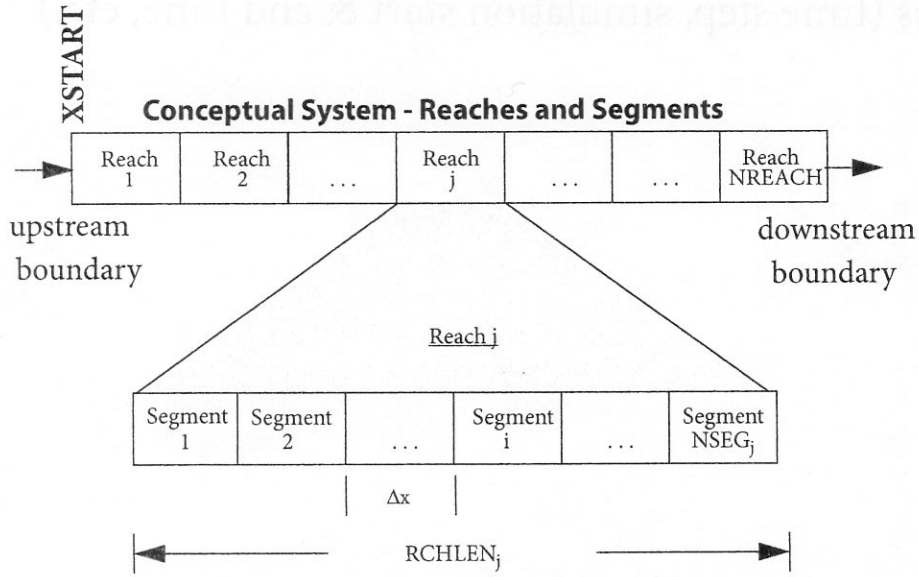
- Define the Modeled System
- Specify the Upstream Boundary Condition
- Specify the Downstream Boundary Condition
- Set transport parameters (D, A_S, α, but NOT A)
- Set miscellaneous variables (time step, simulation start & end time, etc.)

Step 3—Initial OTIS Runs

Parameter File — Define the Modeled System

Uvas Creek—Model data from 105 m as a single reach





Record Types & Fields In Parameter File

Number of Reaches (NREACH) = 1
Length of Reach (RCHLEN) = 300.0 m
Number of Segments w/i reach (NSEG) = 300

Location of U.S. Boundary (XSTART) = 0 m
Number of Print Locations (NPRINT) = 1
Print Location (PRTLOC) = 105.0 m

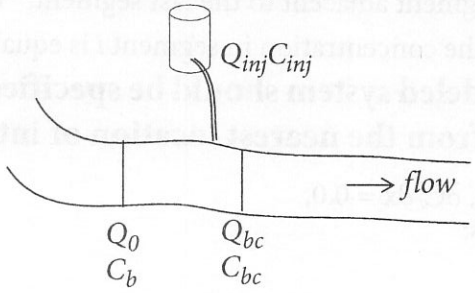
Notes:

1) Δx = 1.0 m (Δx=RCHLEN/NSEG)

2) RCHLEN (300m) > PRTLOC (105m) to keep d.s. boundary away from location of interest (minimize numerical error).

Step 3—Initial OTIS Runs; Parameter File—Upstream Boundary Condition

Continuous Tracer Injection—Mass Balance

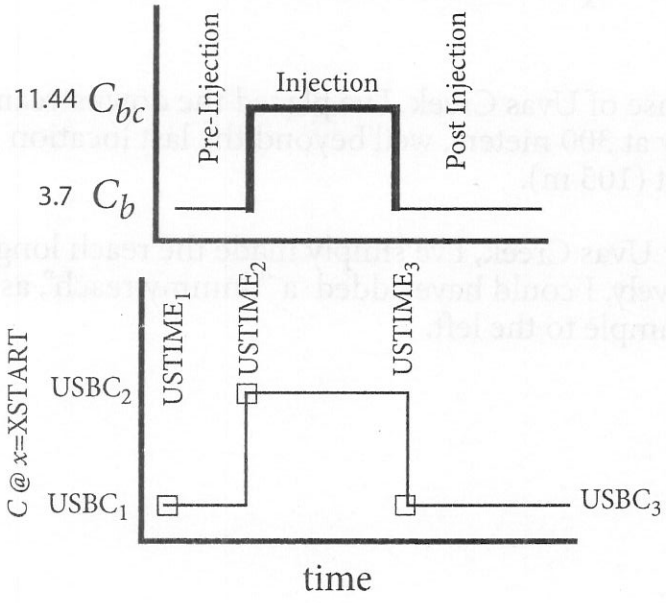


What is the tracer concentration at the upstream boundary, **during the injection**? (see "mass balance is my friend"):

$$C_{bc} = C_b + \frac{Q_{inj}C_{inj}}{Q_{bc}} = 11.44 \text{ mg Cl / L}$$

Concentration at Injection

Concentration-Step b.c. (IBOUND=1)



Record Types & Fields In Parameter File

Number of u.s. boundary conditions (NBOUND) = 3 (pre-injection, injection, post-injection)

Boundary Condition Option (IBOUND) = 1 (Concentration-Step boundary condition)

Pre & Post-Injection b.c. Conc. (USBC1,USBC3) = background tracer conc = 3.7 mg/L

Injection b.c. Conc. (USBC2) = 11.44 mg/L

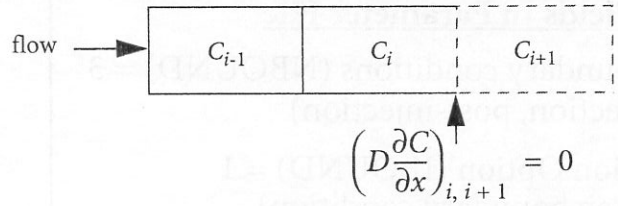
USTIME1 = simulation start time
USTIME2 = injection start time
USTIME3 = injection end time

Note: For our example, the boundary concentration during the injection is equal to the plateau concentration at 105 m (11.44 mg/L)—that's because we don't have inflow (Qbc=Q105). When there's inflow, Cbc > Cplat.

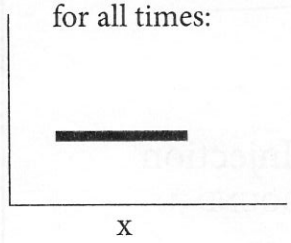
Step 3—Initial OTIS Runs; Parameter File—Downstream Boundary Condition

The downstream boundary condition is defined as a fixed dispersive flux between segments *i* and *i*+1 (DSBOUND), where the subscript *i* refers to the last segment in the modeled system, and *i*+1 refers to a fictitious segment adjacent to the last segment. For *DSBOUND*=0.0, the concentration gradient at the downstream boundary is equal to zero (the concentration in segment *i* is equal to that in segment *i*+1). **Due to this assumption, the length of the last reach in the modeled system should be specified such that the location of the downstream boundary is sufficiently downstream from the nearest location of interest..**

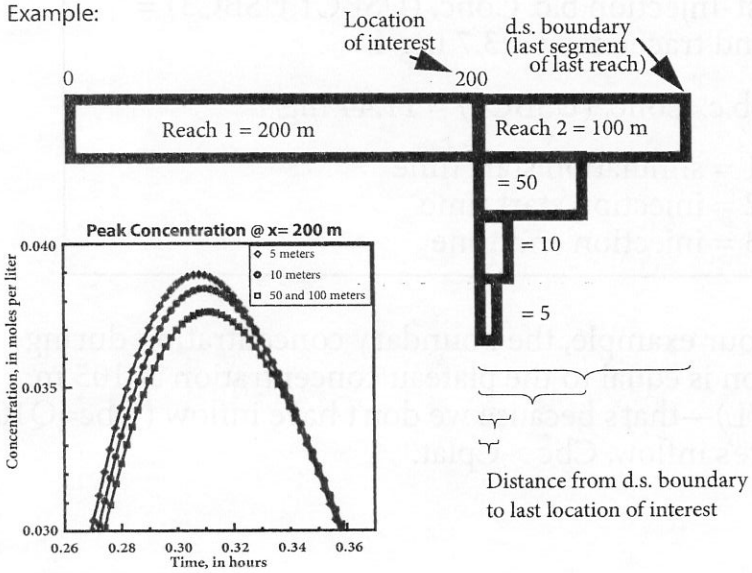
Downstream boundary condition defined in terms of a fixed dispersive flux.



For D > 0.0, $\partial C/\partial x = 0.0$, for all times:



Example:



For the case of Uvas Creek, I've placed the downstream boundary at 300 meters, well beyond the last location of interest (105 m).

Note: For Uvas Creek, I've simply made the reach longer. Alternatively, I could have added a "dummy reach", as in the example to the left.

Step 3—Initial OTIS Runs—Parameter File:

Set transport parameters (D, A_S, α)—

- Tracer profile doesn't suggest much storage (limited tail)
  → As a first cut, model Advection-Dispersion only
  (ALPHA (α) = 0; AREA2 (A_S) to arbitrary value)

  $\alpha = 0$ turns off storage component
- Initial Estimate (guess): DISP (D) = 0.01 m^2/s

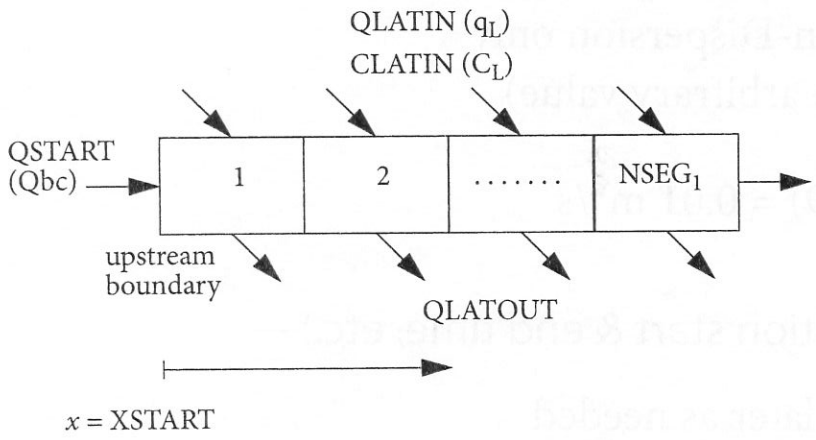
Set miscellaneous variables (time step, simulation start & end time, etc.)—

- Δt = TSTEP = 0.05 hours; revise later as needed
- set print step (PSTEP) equal to TSTEP (output all results)
  only reason to have PSTEP>TSTEP is if you're worried about the size of the output file
- Simulation start and end times (TSTART, TFINAL) define the period you want to simulate.

Step 3—Initial OTIS Runs

Flow File — Flows & Main Channel Cross-Sectional Area

First reach in the stream network



Record Types & Fields — Flow File

QSTART — streamflow at upstream boundary

For Each Reach:

AREA (A) — main channel cross-sectional area

QLATIN (q_{LIN}) — lateral inflow (per unit length):

$$q_{LIN} = \frac{Q_{bottom} - Q_{top}}{\text{reach length}}$$

CLATIN (C_L) = lateral inflow concentration (background concentration)

QLATOUT — lateral outflow (per unit length)

Uvas Creek Example:

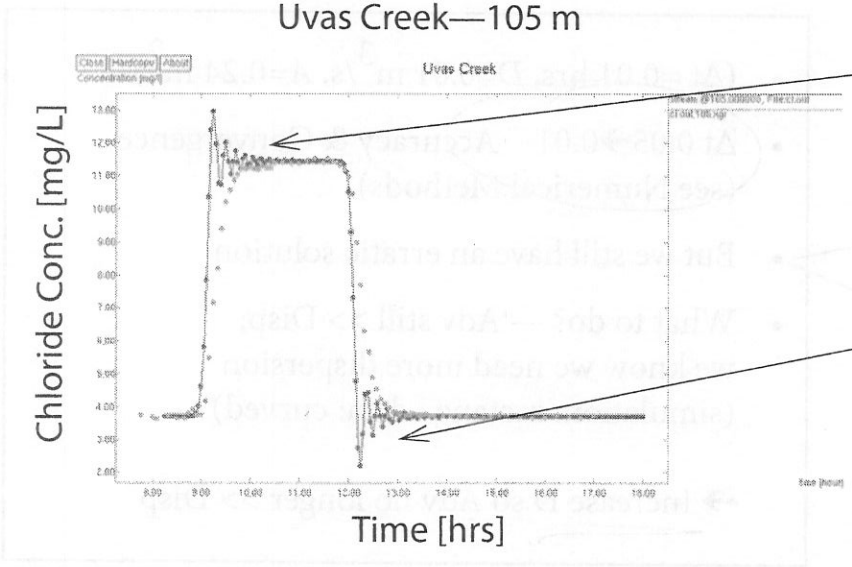
QSTART (Qbc) = 0.0128 (see Step 1)

QLATIN = 0 — no dilution of tracer
→ no inflow (see Step 1)

AREA (A) = 0.24 (see Step 2; initial estimate)

Step 3—Initial OTIS Runs

Uvas Creek Simulation #1 — ($\Delta t = 0.05$ hrs, D=0.01 m^3/s, A=0.24 m^2)

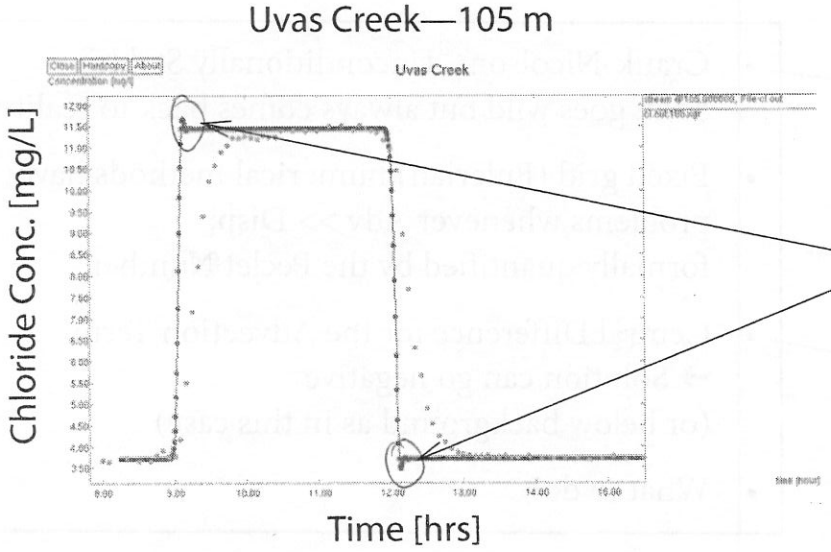


- Crank-Nicolson: "Unconditionally Stable"—
  Soln. goes wild but always comes back to reality
- Fixed grid (Eulerian) numerical methods have problems whenever Adv >> Disp;
  formally quantified by the Peclet Number.
- Central Difference for the Advection Term
  ➔ Solution can go negative
  (or below background as in this case)
- What to do?

data — points only
simulation — lines & points

Step 3—Initial OTIS Runs

Uvas Creek Simulation #2 — Decrease Timestep to Improve Accuracy



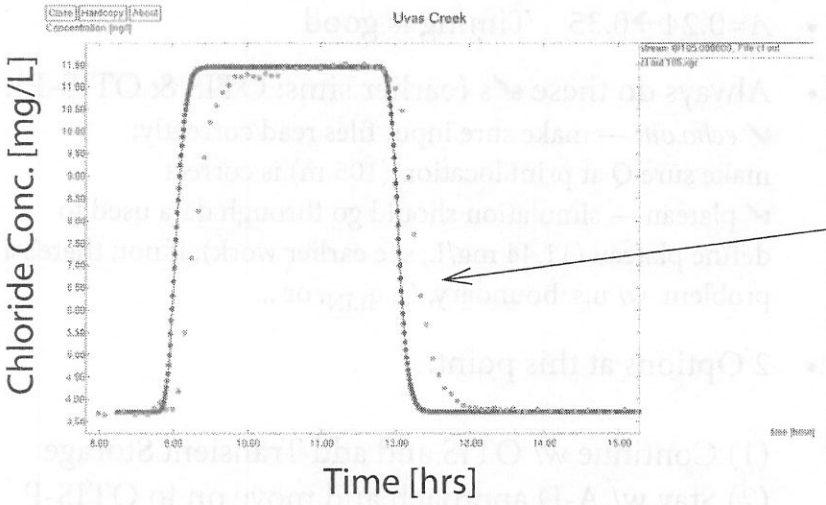
data — points only
simulation — lines & points

- (Δt =0.01 hrs, D=0.01 m^3/s, A=0.24 m^2)
- Δt 0.05→0.01 Accuracy & Convergence (see Numerical Methods)
- But we still have an erratic solution
- What to do? — Adv still >> Disp; we know we need more dispersion (simulation: rectang.; data: curved)

  → Increase D so Adv no longer >> Disp

Step 3—Initial OTIS Runs

Uvas Creek Simulation #3 — Increase *D* to produce curvature & address numerics



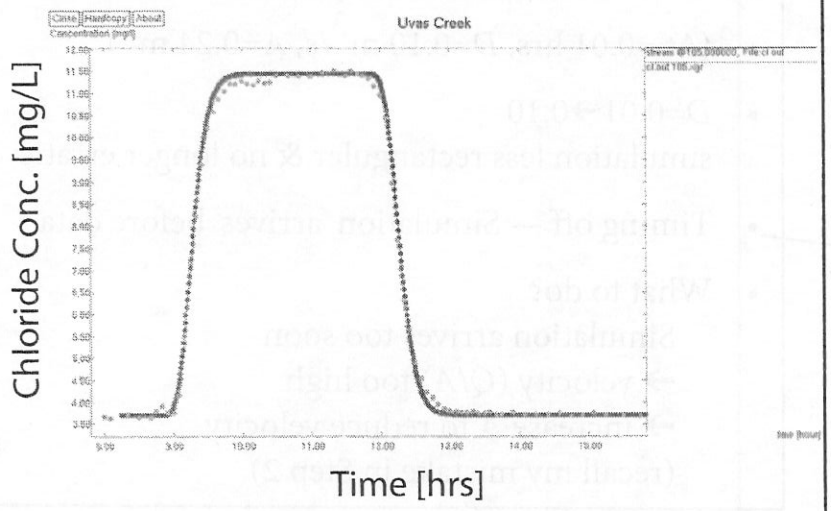
data — points only
simulation — lines & points

- (Δt =0.01 hrs, D=0.10 m^3/s, A=0.24 m^2)
- D=0.01→0.10
  simulation less rectangular & no longer erratic
- Timing off — Simulation 'arrives' before data
- What to do?
  Simulation arrives too soon
  → velocity (Q/A) too high
  → increase A to reduce velocity
  (recall my mistake in Step 2)

Step 3—Initial OTIS Runs

Uvas Creek Simulation #4 — Increase *A* to improve timing

Uvas Creek—105 m



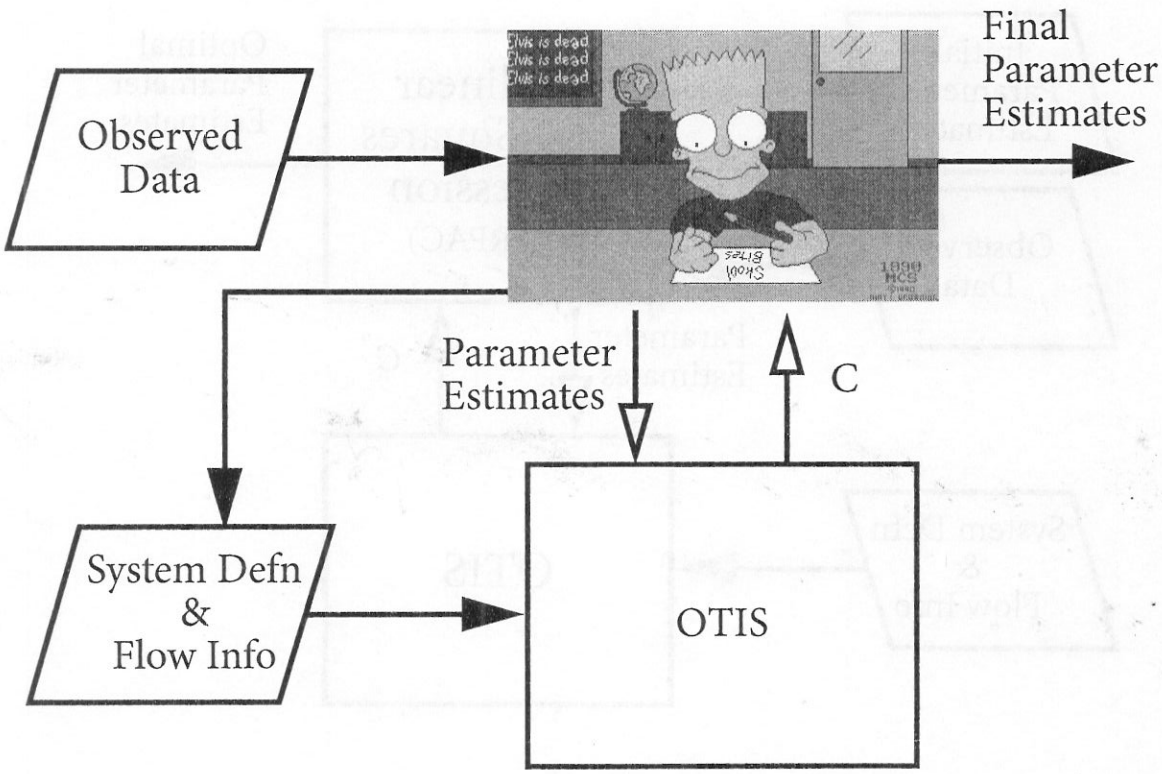
data — points only
simulation — lines & points

- (Δt =0.01 hrs, D=0.10 m^3/s, A=0.35 m^2)
- A=0.24→0.35 Timing is good
- Always do these ✔s (earlier sims; OTIS & OTIS-P):
  ✔ *echo.out* — make sure input files read correctly; make sure Q at print location (105 m) is correct
  ✔ plateau — simulation should go through data used to define plateau (11.44 mg/L; see earlier work); if not, there's a problem w/ u.s. boundary, Q, q_{LIN}, or ....
- 2 Options at this point:

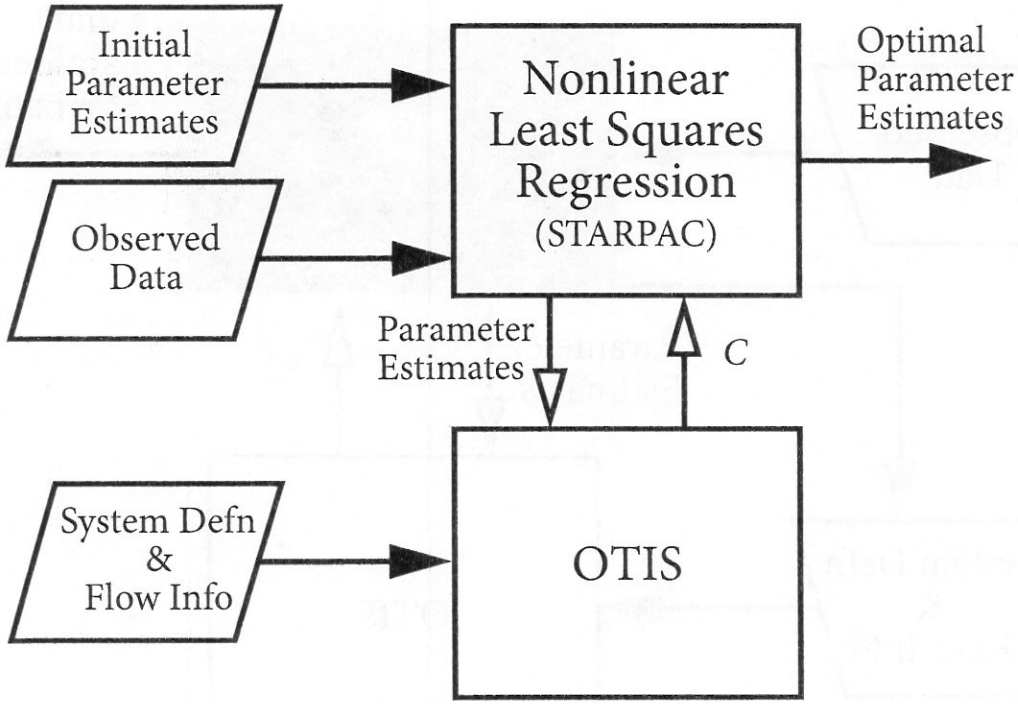
  (1) Continue w/ OTIS and add Transient Storage
  (2) Stay w/ A-D approach and move on to OTIS-P

  Doesn't appear to be much storage → Option #2

Example OTIS Application
Step 4: Automated Parameter Estimation Using OTIS-P — Advection-Dispersion



Example OTIS Application
Step 4: Automated Parameter Estimation Using OTIS-P — Advection-Dispersion



Example OTIS Application

Step 4: Automated Parameter Estimation Using OTIS-P — Advection-Dispersion

Goal: Given a set of observed concentrations, determine a set of parameter values (A, A_S, α, D) that minimizes the residual sum of squares. The residual sum of squares quantifies the discrepancy (error) between the simulation and the observed data.

Given: Observed = Simulated + Error

$$C_k = f(A, D, A_S, \alpha)_k + \varepsilon_k, \qquad k = 1, N$$

Minimize: Residual Sum of Squares:

$$RSS(A, D, A_S, \alpha) = \sum_{k=1}^{N} [C_k - f(t, A, D, A_S, \alpha)_k]^2 = \sum_{k=1}^{N} \varepsilon_k^2$$

where

N — the number of observations

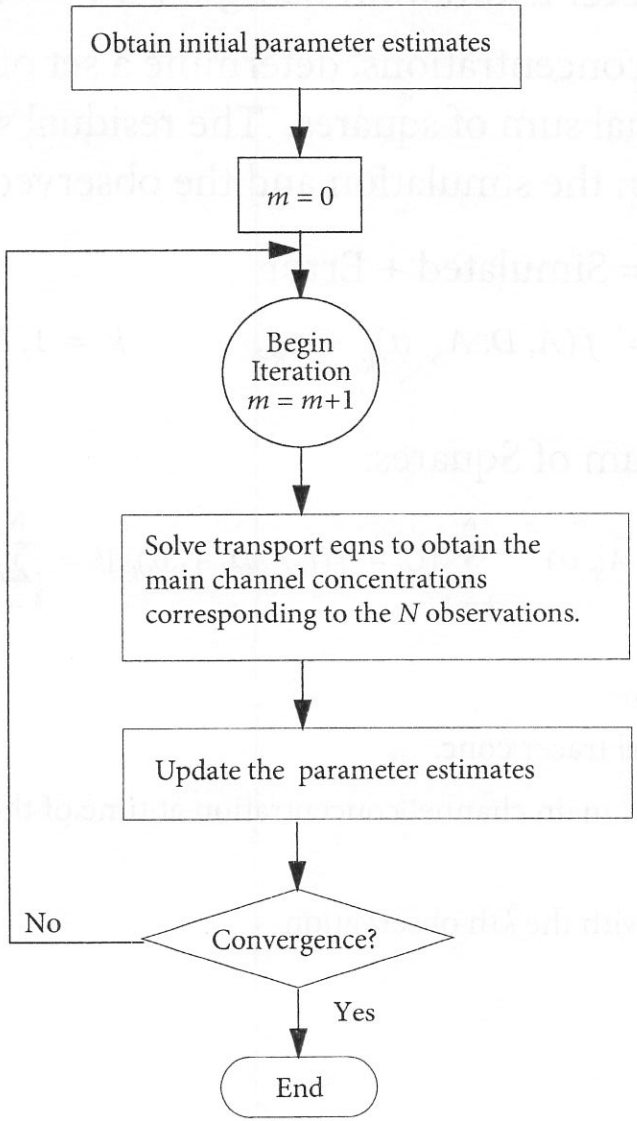
C_k — *kth* observed main channel tracer conc.

$f(t, A, D, A_S, \alpha)$ — simulated OTIS main channel concentration at time of the kth observation

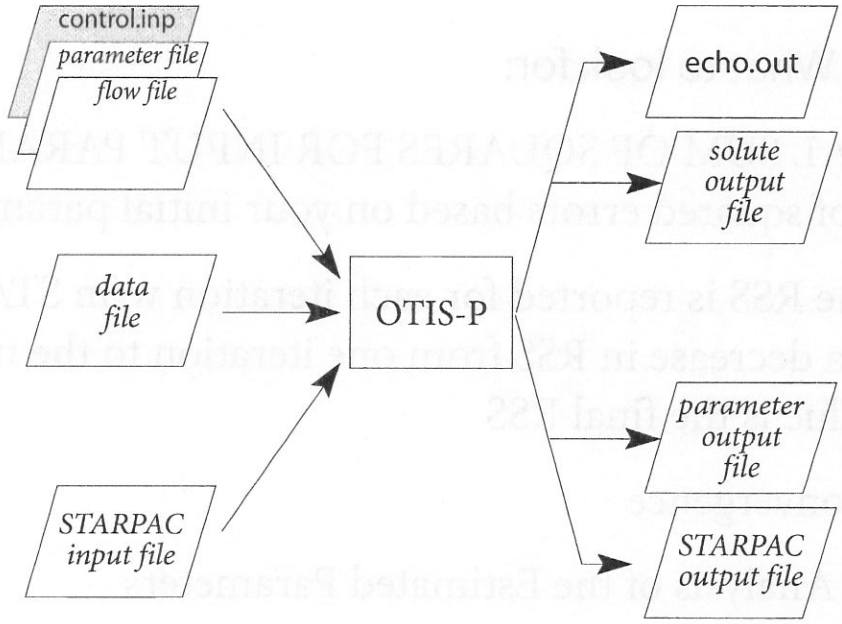
t — time

ε_k — random error associated with the kth observation

Parameter Estimation using Nonlinear Least Squares



Step 4: Automated Parameter Estimation Using OTIS-P — Advection-Dispersion



Input Files

data file — contains observed concentration data corresponding to the print locations; e.g. observed data from 105 m

STARPAC *input file* — used to specify parameters that are to be estimated by OTIS-P and those which are "fixed"; e.g. we'll initially want to estimate D and A, while keeping A_S and α fixed (recall that we're not modeling storage yet—α is fixed at 0.0)

Output Files

parameter output file — contains the final parameter estimates determined using NLS (I rarely look at this one)

STARPAC output file — contains statistical analyses related to the observed data and the NLS procedure

Step 4: Automated Parameter Estimation Using OTIS-P

STARPAC Output File — What to look for:

- "RESIDUAL SUM OF SQUARES FOR INPUT PARAMETER VALUES"
  The sum of squared errors based on your initial parameter estimates.
- RSS — The RSS is reported for each iteration w/in STARPAC —
  you'll see a decrease in RSS from one iteration to the next;
  the last value is the final RSS
- Type of Convergence
- Statistical Analysis of the Estimated Parameters

Step 4: Automated Parameter Estimation Using OTIS-P

STARPAC Output File — Types of Convergence:

- RSS Convergence — generally a good sign
- Parameter Convergence — generally a good sign
- Singular Convergence — generally a bad sign
- False Convergence — generally a bad sign
- Singular and/or False Convergence: These types of convergence are discussed in the STARPAC User's Guide (Donaldson and Tryon, 1990):

"Singular convergence means that the model contains too many parameters, at least near the solution, while false convergence can indicate that either STOPSS or STOPP is set too small for the accuracy to which the model and its derivatives are being computed or that there is an error or discontinuity in the derivative."

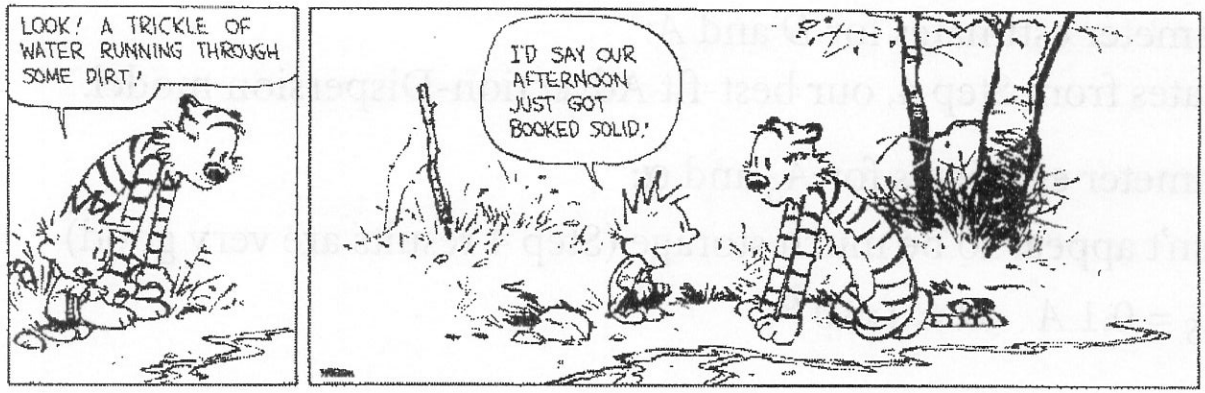
My Take: There's not enough "information" in the data to estimate the parameters; i.e. you're trying to estimate transient storage, but you don't have enough resolution in the arrival and/or departure data. This often leads to "parameter non-uniqueness" — different sets of parameters produce equally good (bad) fits. Sometimes you can get around it be reducing the number of estimated parameters, i.e. fixing D rather than estimating it — other times you just have bad data, and there's not much you can do.

Step 4: Automated Parameter Estimation Using OTIS-P

General Procedure for Using OTIS-P — the Multi-Pass Approach:

- I generally do a series of OTIS-P runs — I refer to these runs as "Passes"; See Application 5 in the documentation for an example
- For a given pass: use the final parameter estimates from the previous pass as initial estimates for the current pass (parameter estimates are taken from the STARPAC output file and used as initial estimates in revised parameter and/or flow files)
- This multi-pass approach sometimes eliminates False and Singular convergence.
- The multi-pass approach concludes when the final parameter values and residual sum of squares are invariant from one pass to the next

Example OTIS Application
Step 5: Using OTIS-P, add Storage — Advection-Dispersion-Storage



Example OTIS Application
Step 5: Using OTIS-P, add Storage — Advection-Dispersion-Storage

- STARPAC Input File: D, A, A_S and α are now all estimated parameters.
- Initial parameter estimates for D and A:
  final estimates from Step 4, our best-fit Advection-Dispersion model.
- Initial parameter estimates for A_S and α:
  There doesn't appear to be much storage (Step 4 results are very good)
  → try: $A_S = 0.1\ A$ $\alpha = 1 \times 10^{-6}$

Step 5: Automated Parameter Estimation Using OTIS-P
Uvas Creek, Advection-Dispersion-Storage — Results of Multi-Pass Approach

	Convergence Type	D [m^2/sec]		A [m^2]		A_S [m^2]		α [/sec]		Input RSS	Final RSS
		Value	Ratio	Value	Ratio	Value	Ratio	Value	Ratio		
Initial Est. (Step 4 AD & guesses)		0.130		0.352		0.0350		1.00E-6			
Pass 1	Singular	0.123	—	0.351	—	0.0314	—	5.56e-6	—	**1.418**	1.112
Pass 2	RSS	0.117	22.9	0.349	238.5	0.0493	2.33	8.57e-6	5.16	1.112	1.043
Pass 3	Param & RSS	0.117	22.9	0.349	238.5	0.0493	2.33	8.57e-6	5.16	**1.043**	1.043

- input RSS for Pass 1 (1.418) < final RSS for AD model (1.545, Step 4)
  → our initial guesses for A_S and α improve the fit—our AD-storage model is 'better' than the A-D model, and we haven't even run OTIS-P yet. This is sheer luck. In many cases the initial guesses make the fit worse—if its a lot worse, you may need to lower the values of the initial estimates (or do trial-&-error OTIS runs to develop estimates)
- Multi-pass approach eliminates Singular convergence problem; "good" convergence.
- Pass 2: Initial RSS equal final RSS from Pass 1; Pass 3: Initial RSS equal final RSS from Pass 2 ✔
- No change from Pass 2 to Pass 3 → Advection-dispersion-storage model is done

Step 5: Automated Parameter Estimation Using OTIS-P
Uvas Creek—Final Advection-Dispersion-Storage Model

D [m^2/sec]				A [m^2]				A_S [m^2]				α [/sec]			
Lower 95%	Value	Upper 95%	Ratio	Lower 95%	Value	Upper 95%	Ratio	Lower 95%	Value	Upper 95%	Ratio	Lower 95%	Value	Upper 95%	Ratio
0.11	0.12	0.13	23	0.35	0.35	0.35	239	0.007	0.049	0.092	2.3	5.3e-6	8.6e-6	1.2e-5	5.2

- In general, these results are very good — "Good" (P&RSS) convergence & high ratios
- A has the highest ratio—this is typically the case
  —A well defined by the arrival & departure (see Sensitivity diagram presented earlier and our traveltime calculation)
- Ratios for A_S & α lower than A & D → more uncertainty
- A_S uncertainty: lower 95% value = 1/7 estimated value
  upper 95% value = 1.9 * estimated value

Step 5: Automated Parameter Estimation Using OTIS-P
Uvas Creek—Final Advection-Dispersion-Storage Model

Model-derived Transport Characteristics:

Note: velocity & traveltime should be calculated using the model-derived estimate of main-channel cross sectional area (*A*). Many papers present velocity estimates based on 'time to 1/2 plateau', velocity of the leading edge/peak, moment analysis, etc. — but the modeling provides better estimates of velocity & traveltime due to the fact that *A* is part of the advection term, and that term is fit to the data.

Velocity

$U = Q/A = 0.0128 m^3/s / 0.35 m^2 = 0.04$ m /s

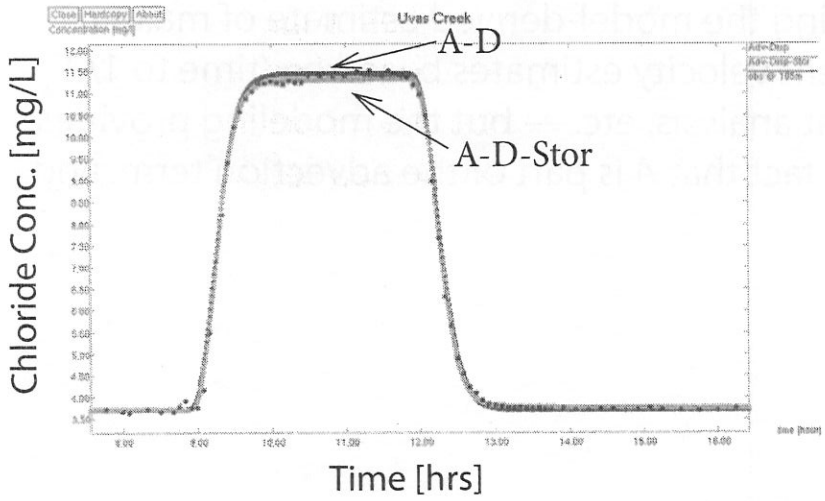
Traveltime

t_t = distance/velocity = 105 m / 0.0366 m/s = 2870 seconds

Step 5: Uvas Creek—Final Advection-Dispersion-Storage Model

How important is transient storage?l

Uvas Creek—105 m



An Apparent Contradiction

<u>Evidence for storage:</u>

- A-D-Stor model—Good convergence, good ratios
- $A_S / A = 0.049 / 0.35 = 0.14$
  not bad—relative to the 53 data sets summarized in Runkel (2002), this would rank ~30th

→ Transient storage is important in the first 105 m of Uvas Creek!

<u>BUT...</u>

- observed data—does there appear to be a storage effect ? (the emperor has no tail!)
- fits very similar—compare plot at left to Uvas plot from 433m (earlier in notes)—(w/ the exception of a small shoulder/tail effect, the A-D model does well)

Step 5: Uvas Creek—Final Advection-Dispersion-Storage Model

How important is transient storage?

Mean Travel Time Due To Storage
(alternate way to express A_S/A)

$$F_{mean} = \frac{A_S}{(A + A_S)} = 0.124$$ (expressed as a fraction; multiply by 100 to express as %)

12.4 % of mean traveltime is attributed to storage

→ just as before, analysis based on A_S & A indicates that storage is important

Median Travel Time Due To Storage (Equation 14, Runkel, 2002)
(based on tracer mass; considers interaction of A_S, A, α, and velocity)

$$F_{med}^{L=105m} \cong \left(1 - e^{-L\frac{\alpha}{(Q/A)}}\right)\frac{A_S}{(A + A_S)} = 0.003$$ (expressed as a fraction; multiply by 100 to express as %)

0.3 % of median traveltime is attributed to storage

During the 2870 sec for traveltime from the injection to 105m, 9 seconds are spent in storage

→ **Transient storage is NOT important**—Bencala and Walters were right (they set α=0 for the first 105m).

Step 5: Uvas Creek—Final Advection-Dispersion-Storage Model

Some final words on Interpretation:

Fmed200 —Standard for Inter-Reach, Inter-Stream Comparisons (Runkel, 2002)

$$F_{med}^{L=200m} \cong \left(1 - e^{-L\frac{\alpha}{(Q/A)}}\right)\frac{A_S}{(A + A_S)} = 0.006$$ (expressed as a fraction; multiply by 100 to express as %)

If the reach was 200 m long, 0.6 % of median traveltime would be spent in storage

Damkohler Number (Wagner & Harvey, 1997)

$$DaI = \frac{L\alpha\left(\frac{A + A_S}{A_S}\right)}{Q/A} = 0.2$$

DaI << 1; limited interaction between tracer and storage zone—exchange is not "identifiable" (Harvey & Wagner, 2000).

My take: Storage parameters are identifiable w/ high quality data set used here—they're just not important (as per Fmed).

Many similarities w/ Fmed—appears to be somewhat redundant

Step 4: Automated Parameter Estimation Using OTIS-P

Statistical Analysis (available only for RSS and/or Parameter Convergence):

- The STARPAC output file includes:
  Parameter Estimate, Standard Deviation of Estimate, The Ratio, 95% Conf. Interval
- Ratio = Estimate/Standard Deviation; i.e. inverse of the coefficient of variation
  → You want high ratios (low variation, i.e. low uncertainty)
- Ratio < 1; the standard deviation is greater than the parameter estimate itself
  → the parameter estimate is meaningless; if you don't believe me, take a look at the 95% confidence interval — the estimate is highly uncertain)
- Publications: Report the estimate and the ratio; use 95% confidence interval to put error bars on your estimates.
- RSS is relative — the magnitude depends on how much data you have, your units, etc.
  → don't report it unless your comparing "Apples & Apples" (e.g. the simulation w/ just Advection-Dispersion has an RSS of 3.94, the simulation w/ Advection-Dispersion-Storage has an RSS of 3.01; adding storage therefore improves the fit)

Step 4: Automated Parameter Estimation Using OTIS-P

Back to Uvas Creek — Try to improve upon our manual OTIS fit

- Initial parameter estimates for D and A: Use the parameter and flow files from our last OTIS run (Simulation 4 from Step 3).
- STARPAC Input File: D and A estimated parameters; everything else is fixed.
- Data File: Contains observed data from 105 m

Step 4: Automated Parameter Estimation Using OTIS-P

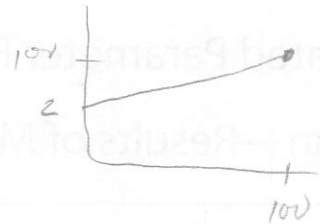
Uvas Creek, Adection-Dispersion—Results of Multi-Pass Approach:

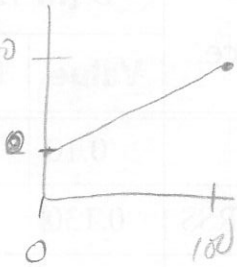
	Type of Convergence	D [m^2/sec]		A [m^2]		RSS, Input Parameter values	RSS
		Value	**Ratio**	**Value**	**Ratio**		
Initial est. (OTIS,Sim #4)		0.10		0.35			
Pass 1	Parameter & RSS	0.130	22.1	0.352	240.7	2.16	1.545
Pass 2	Parameter & RSS	0.130	22.1	0.352	240.7	1.545	1.545

these two values must be same

- OTIS-P impoves a the fit (RSS decreases relative to OTIS Simulation #4); no surprise, that's the whole idea.
- "Good" Convergence, high ratios.
- Pass 2: Initial RSS equals final RSS from Pass 1 (this should always be the case for single reach problems)
  → we didn't mess up when we took final pass 1 estimates from the STARPAC output file for use as initial estimates in pass 2. ✔
- No change from Pass 1 to Pass 2 → Advection-dispersion model is finalized

Ratio value generally larger for A compared to D
Ratio = 240.7 = Estimate is 240 x greater than Std Dev.


RWT volts
10V
2
10V


100
0
0
100